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CHROMATE CONVERSION

THE COMPLETE TRAINING MANUAL

CONVERSION COATING · CHROMATE ON ZINC · CR VI

A full training course for the brand-new operator — from "what even is a passivate?" to a signed certificate of completion. The classic clear, yellow, olive-drab, and black ***chromates*** *that finish freshly zinc-plated parts: the self-healing film that turns shiny zinc into a corrosion-fighting finish. No electricity, no plating current — just chemistry. It also teaches the most hazardous chemistry in the conversion-coating family:* ***hexavalent chromium.*** *Assumes you know nothing. Teaches you everything.*

01 RECEIVE ▸ 02 ACTIVATE ▸ 03 RINSE ▸ 04 CHROMATE

05 RINSE ▸ 06 SEAL ▸ 07 DRY·COOL ▸ 08 INSPECT

OPERATOR TRAINING & CERTIFICATION

EDITION 1.0 · 2026 · BEGINNER-FRIENDLY STYLE

PART ONE — FRONT MATTER & FUNDAMENTALS

START HERE

WELCOME ABOARD

IF YOU DON'T KNOW A PASSIVATE FROM A PAPERWEIGHT, YOU'RE IN EXACTLY THE RIGHT PLACE — AND HERE THAT'S ABOUT YOUR HEALTH, NOT JUST QUALITY

Welcome. If you're holding this manual, you're about to learn how to run — or at least understand — a **chromate conversion line**: the quick chemical dip that goes on **freshly zinc-plated parts** to turn bright, bare zinc into a corrosion-resistant, colored, paint-ready finish. Maybe you started this week; maybe you've been pulling baskets off the zinc line for a year and want to know *why* that last yellow dip matters. Either way, you're in the right place.

Here is the most important thing we can tell you up front, and we will say it more than once: **the classic chromate bath runs on hexavalent chromium — Cr(VI) — a confirmed human carcinogen.** It is the single most hazardous chemistry in the whole conversion-coating family. A trained, careful operator who respects it can work with it safely for an entire career — an untrained one cannot. The second thing: **this manual assumes you know nothing about chemistry, zinc plating, or how a "passivate" works.** That's not an insult — it's a promise. Every term gets defined the first time we use it, in the spirit of the *plain-language* guides: friendly, plain-spoken, and patient.

★ **REMEMBER**

You do not need to memorize this manual. You need to *understand* it. Understanding sticks. Memorizing fades by Friday — and on a Cr(VI) line, the understanding is what keeps you healthy.

⚠ **HEADS-UP**

One big idea before we start: **a chromate line has no electricity running through the parts.** There's no rectifier, no anode, no cathode — the coating forms because the zinc surface and the chromate solution *react with each other.* That removes the shock-at-the-part hazard. It does **not** make the line safe: you are still working with a **carcinogenic hexavalent-chromium bath and its mist**, plus acidic activation dips, hot drying air, and slick wet floors. Respect all of it.

• HOW TO USE THIS MANUAL

The manual follows the **line itself**, in the order a part travels — receive, activate, rinse, chromate, rinse, seal, dry, inspect. Reading it in order matters: the order of a finishing line is never random, and neither is the order of this book.

💡 **TIP**

Read Part One all the way through *before* any individual station chapter — the station chapters assume you already understand the big picture (especially the Cr(VI) hazard and self-healing) that Part One gives you.

THE ICONS — YOUR FRIENDLY MARGIN HELPERS

💡 **TIP**

A shortcut or trick of the trade — the kind of thing a 20-year veteran would lean over and tell you.

★ **REMEMBER**

A core idea worth locking in. If you forget everything else on the page, don't forget these — they're the load-bearing ideas.

⚠ **HEADS-UP**

A safety warning. Read every single one. On a chromate line these keep your lungs, skin, eyes, and long-term health intact. We use them *very* liberally — Cr(VI) earns it.

⚙ **TECHNICAL STUFF**

Deeper chemistry for the curious. Optional — skip it and you can still run the line.

☀ **FUN FACT**

A bit of trivia to make the science stick (and give you something to say at lunch).

TWO THINGS YOU'LL SEE AT EVERY STATION

Each station chapter ends with a **Teach-Back** checklist (paired with a "Top Mistakes" card) — the things you must say out loud, unprompted, before you're cleared — and a bordered **Sign-Off** box your *trainer initials* once you've demonstrated the skill safely. Teach-Back is your study guide; Sign-Off is the gate.

FIND YOUR WAY

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READ FRONT-TO-BACK THE FIRST TIME · KEEP ON THE SHELF FOREVER AFTER

PART ONE — FRONT MATTER & FUNDAMENTALS

PART TWO — THE LINE, STATION BY STATION (STATIONS 01–08)

PART THREE — ACROSS THE LINE

PART FOUR — BACK MATTER & CERTIFICATION

★ **REMEMBER**

This manual is a training and reference tool — **not** a substitute for your facility's written procedures, your supplier's TDS/SDS, your shop's OSHA Cr(VI) compliance program, or your supervisor's instructions. When this manual and your shop's official procedure disagree, **follow your shop's official procedure and ask your supervisor.**

WHAT YOU'LL BE ABLE TO DO

LEARNING OBJECTIVES

MASTER THIS FOUNDATION AND EVERY STATION CHAPTER CLICKS INTO PLACE

By the time you finish this manual, you should be able to confidently:

1. **Explain what a conversion coating is** in plain language — and how it differs from electroplating and anodizing (chemistry alone, *no electric current*).
2. **Describe what a chromate (passivate) is and what it does** — a thin, gel-like chromium/zinc film grown on freshly zinc-plated parts that dramatically boosts corrosion resistance, is **self-healing**, adds color, and serves as a paint base.
3. **State that the chromate is the *finishing step after zinc plating*** — the "passivate" the zinc manuals hand off to — and name the substrates it works on (zinc, zinc-nickel, zinc-iron, cadmium).
4. **Read the classic color palette** — clear/blue, yellow/gold iridescent, olive-drab/bronze, and black — and know that heavier/darker generally means more corrosion protection, while clear means the least.
5. **State the central safety fact of this line:** the classic bath is **hexavalent chromium (Cr VI), a confirmed human carcinogen**, and name the controls that keep you safe (ventilation, mist suppression, respiratory protection, hygiene, $Cr^{6+} \rightarrow Cr^{3+}$ waste treatment).
6. **Explain the regulatory driver** — that **RoHS, ELV, and REACH restrict hexavalent chromium**, which is why most of the industry has moved to trivalent (Cr III) passivates (our next manual).
7. **Name the stations of the line in order** and explain *why* the order can't be rearranged.
8. **Apply the single most important operator process lesson** — *do not dry the fresh film hot* — and explain what over-drying does to a soft gel chromate.
9. **Spot trouble early** — no color, powdery/rub-off film, poor salt spray, dried-out/crazed film — and talk the language with your lead, your chemist, and your supplier.

> ★ **REMEMBER**
> You are not expected to hit all of these on day one. Surface finishing is a craft. But the *safety* objectives (#5) are not optional and not gradual — you must have those before you work the line at all.

• THE EIGHT-STATION PROCESS AT A GLANCE

The whole chromate line in one strip. Don't memorize it yet — just feel the rhythm: **Receive → Activate → Rinse → CHROMATE → Rinse → Seal → Dry (cool) → Inspect.**

01	02	03	04	05	06	07	08
RECEIVE	**ACTIVATE**	**RINSE**	**CHROMATE**	**RINSE**	**SEAL**	**DRY**	**INSPECT**
zinc in	nitric dip	overflow	pH 1.5-2.5, Cr VI	cool, gentle	chrome/non	≤60°C cool	color & rub

> ★ **REMEMBER**
> Notice the pattern: **a rinse follows each working stage**, and the **dry-off is deliberately *cool*, not hot.** That last idea makes this line very different from the iron phosphate and hard chrome lines, where heat is welcome. Here, a hot oven would *destroy* the coating you just grew.

> ⚠ **HEADS-UP**
> This eight-station layout is the *teaching* version. Your shop's real line may add or combine steps (e.g., a separate desmut, a dragout-recovery rinse, multiple seals). Always run *your* line's actual posted sequence and parameters.

CHAPTER 1

WHAT IS A CONVERSION COATING?

THE TRUE BEGINNER'S PRIMER — SO FAR BACK WE DON'T EVEN ASSUME YOU KNOW WHAT "FINISHING" MEANS

THE ONE-SENTENCE VERSION

A conversion coating is a protective film you grow on a metal surface by making the metal itself react with a chemical solution — turning the top layer of the metal *into* the coating. That's the whole idea. Now let's unpack it, because the word "conversion" is doing a lot of work.

THREE COUSINS: PLATING, ANODIZING, AND CONVERSION COATING

If you've seen our other manuals, you've met two other finishing families. It's worth lining all three up side by side, because the differences are the key to understanding chromates.

- **Electroplating** *adds* a layer of a *different* metal on top of your part, using **electricity**. (Zinc plating — the step right *before* this one — is electroplating: electricity drives dissolved zinc out of a bath and onto the steel.)
- **Anodizing** *converts* the part's *own* surface into a thick, hard oxide — but it still uses **electricity** to drive the reaction, and only on metals like aluminum.
- **Conversion coating** (this manual) *converts* the part's own surface into a thin protective film using **only chemistry — no electricity at all.** You dip the part, the surface reacts with the solution, and a new compound forms right where the bare metal used to be.

★ **REMEMBER**

Lock in the difference: **Plating adds a new metal with electricity. Anodizing converts the surface with electricity. A conversion coating converts the surface with *chemistry alone* — no rectifier, no anode, no cathode, no current.** If you ever go looking for the "power supply" on a chromate line, you won't find one feeding the parts. That's not a missing piece — that's the whole point.

WHERE THIS ONE FITS: AFTER PLATING, NOT INSTEAD OF IT

Here's the twist that makes chromate special among conversion coatings: **the surface it converts is itself an electroplated layer.** A chromate doesn't react with steel — it reacts with the **zinc** (or zinc-alloy, or cadmium) that was just plated *onto* the steel. So a chromate line is almost always the **tail end of a zinc-plating line**: plate the zinc, rinse it, then passivate it. The zinc protects the steel; the chromate protects the zinc.

☀ **FUN FACT**

Chromate conversion coatings go back to the 1920s–1930s, and the U.S. military leaned on them heavily in World War II — the olive-drab chromate on countless steel fittings and hardware is the same family you'll run today. For nearly a century, "zinc plate and yellow chromate" was *the* default low-cost corrosion finish for steel hardware.

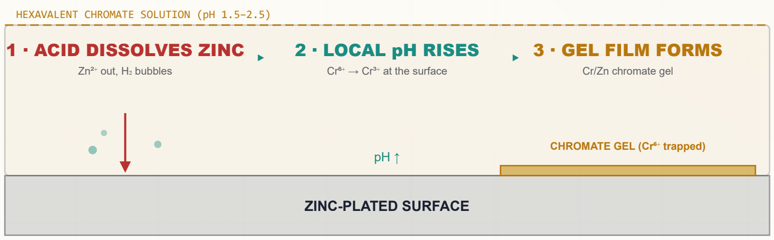
HOW A CONVERSION COATING FORMS — NO CURRENT REQUIRED

So if there's no electricity pushing metal around, what makes the coating appear? The metal and the solution do it themselves. Here's the plain-language version for a chromate on zinc:

1. The chromate solution is **strongly acidic** (pH around 1.5–2.5). When a freshly zinc-plated part touches it, the acid begins to **dissolve a little of the zinc surface** — zinc goes into solution and tiny hydrogen bubbles form. (You'll often see fine bubbling on parts in the dip — that's the reaction working.)
2. That dissolving reaction does two things at once: it **uses up acid right at the surface (so the local pH climbs)**, and it **hands electrons to the dissolved hexavalent chromium**, reducing some of it from **Cr^{6+} to Cr^{3+}**.
3. When the local acidity drops past a tipping point, the trivalent chromium, zinc, and chromate can no longer stay dissolved at the surface — they **precipitate** as a thin, gel-like film of **hydrated chromium and zinc chromates/oxides** right onto the part.
4. Crucially, that soft gel film **traps some of the still-soluble hexavalent chromium inside it.** Hold that thought — it is the secret to *self-healing* (Chapter 2).

The result is an **integral** coating — chemically grown into the zinc surface, not glued on top — that is dramatically more corrosion-resistant than bare zinc, takes color, and gives paint something to grip.

HOW THE FILM FORMS (NO ELECTRICAL CELL)



No electricity — the zinc and the solution do it themselves.

TECHNICAL STUFF

Roughly: acid attacks zinc ($Zn + 2H^+ \rightarrow Zn^{2+} + H_2\uparrow$), consuming acid and raising the interfacial pH. The freed electrons reduce hexavalent chromium ($Cr^{6+} + 3e^- \rightarrow Cr^{3+}$), and the rising pH precipitates a gel of hydrated **Cr(III) oxide/hydroxide + zinc chromate**, with **Cr(VI) entrained** in the gel. So a chromate film is a *mixture*: a trivalent-chromium backbone holding a reservoir of soluble hexavalent chromium. That mixed nature is the whole story of this coating.

HEADS-UP

"No electricity at the part" removes the shock-at-the-tank hazard — but the chromate bath itself is a **carcinogenic hexavalent-chromium solution**, and the line's heaters, pumps, hoists, and dryer are still serious electrical equipment. **Any maintenance still requires Lockout/Tagout (LOTO).** No current through the parts does not mean no hazard in the building.

CHAPTER 2

WHAT IS A CHROMATE (PASSIVATE)?

THE THIN, SELF-HEALING FILM THAT FINISHES A ZINC-PLATED PART — CLEAR, YELLOW, OLIVE-DRAB, OR BLACK

A CHROMATE’S BIG JOBS: PROTECT, HEAL, COLOR, AND BOND

Here’s the headline, and it’s one of the most important ideas in this manual: **a chromate (also called a “passivate” or “conversion coating”) is the finishing step that takes a freshly zinc-plated part and makes it far more corrosion-resistant — and uniquely, the coating can *heal its own scratches.*** It does four jobs at once:

1. **Corrosion protection.** Bare electroplated zinc is shiny but corrodes quickly, forming powdery white “white rust.” A chromate film slows that dramatically — multiplying the salt-spray life of the part many times over.
2. **Self-healing.** This is the signature property of *hexavalent* chromate, and nothing else in the conversion-coating family does it as well. We’ll devote a whole section to it.
3. **Color.** Clear/blue, iridescent yellow/gold, olive-drab/bronze, or black — both for appearance and as a visual signal of *how much* protection is present.
4. **Paint/adhesive base.** The slightly textured, chemically stable film gives paint, powder, and adhesives a better grip than bright bare zinc.

> ★ **REMEMBER**
> A chromate is **the passivate the zinc line hands off to.** Zinc plating protects the steel; the chromate protects the *zinc* and gives the part its final color and corrosion rating. “Zinc plate” and “chromate” are two halves of one finish — almost nobody ships bright, un-passivated zinc.

THE SIGNATURE TRICK: SELF-HEALING

Why is a chromate so much better than the bare zinc underneath? Partly because it’s a barrier — but mostly because of **self-healing.** Remember from Chapter 1 that the gel film traps a reservoir of soluble **hexavalent chromium** inside it. When the film is later **scratched, nicked, or cut** and that bare spot gets wet, the trapped Cr^{6+} **leaches out of the surrounding film, migrates to the exposed zinc, and re-passivates it** — chemically re-forming protective chromate right over the damage. The coating, in effect, **patches its own wounds.**

> ★ **REMEMBER**
> **Self-healing = the soluble hexavalent chromium in the film leaching out to re-protect a scratch.** This is *the* reason hex chromate has been so valued for so long, and — because it depends on mobile Cr(VI) — it is also exactly the property that the modern trivalent (Cr III) passivates *cannot* fully reproduce. Keep that trade-off in mind; it’s the heart of the hex-vs-trivalent story (Chapter 3 and Part Three).

> ☀ **FUN FACT**
> That self-healing behavior is sometimes called “active corrosion protection,” to distinguish it from a plain “passive” barrier like paint. A scratched painted panel just rusts at the scratch; a scratched hex-chromate panel can actually re-heal the scratch. It’s one of the few coatings that fights back.

THE CLASSIC COLOR PALETTE — AND WHY COLOR TELLS YOU PROTECTION

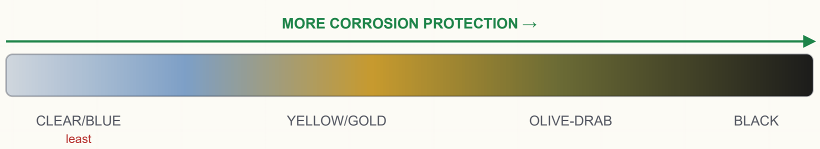
A chromate's color comes from a mix of **film thickness**, the **amount of trapped hexavalent chromium**, and (for some) **added dyes.** As a rough rule on the classic decorative chromates: **the thicker and darker the film, the more chromium it holds, and the more corrosion protection it gives.** Clear is the thinnest and least protective; olive-drab is the heaviest and most protective.

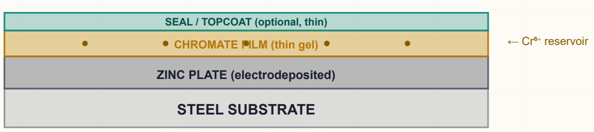
COLOR / TYPE	LOOK	RELATIVE FILM WEIGHT	RELATIVE PROTECTION	TYPICAL USE
Clear / blue-bright	Nearly colorless, faint blue iridescence	Lightest	**Lowest** of the classics	Bright, decorative hardware; modest corrosion demand
Yellow / gold iridescent	Shimmering yellow-gold, rainbow iridescence	Medium	**High** — the workhorse	General-purpose corrosion chromate on zinc hardware/fasteners
Olive-drab / bronze	Dull olive-green to brown	Heaviest	**Highest** of the classics	Military and heavy-duty industrial hardware
Black	Deep black (needs special additives)	Medium–heavy	Good (spec-dependent)	Decorative/functional black; optics, firearms, fasteners

★ REMEMBER

Two ideas to lock in. **(1) Color is a *signal of protection*, not just appearance** — on the classic chromates, clear/blue is the lightest protection and olive-drab the heaviest, with yellow/gold the high-protection workhorse in between. **(2) Color comes from film thickness + trapped Cr(VI) + dyes**, which is why a *faded* or *washed-out* yellow can mean a thin, under-protective film. Always confirm the **exact color/type your job spec calls for.**

READING A CHROMATE BY COLOR · AND WHERE THE FILM SITS





The chromate is the self-healing skin over the zinc.

⚙ TECHNICAL STUFF

The iridescent yellow color is largely the *inherent color of the entrapped hexavalent chromium itself*, layered with thin-film interference (the same soap-bubble physics that colors an oil slick). Olive-drab and black rely on additional chemistry/additives in the bath. One practical consequence: because the yellow color *is* partly the Cr(VI) reservoir, a **bright, full yellow is a visual cue that the self-healing reservoir is present**, and a pale/clear-ish yellow can indicate it was leached or over-rinsed out.

• WHERE CHROMATES WORK — THE SUBSTRATES

A chromate needs a reactive, freshly plated metal surface to convert. It works on:

- **Electroplated zinc** — the classic and most common substrate (this manual's focus).
- **Zinc-alloy plating** — **zinc-nickel, zinc-iron, zinc-cobalt** — increasingly common for higher corrosion performance; these take chromates/passivates too (often with alloy-specific chemistries).
- **Cadmium** — historically chromated the same way (cadmium itself is now restricted in many applications).

> ⚠ **HEADS-UP**
>
> **A different metal needs a different conversion coating.** The chromate that finishes **aluminum** (the classic "chem film" / "Alodine," covered by **MIL-DTL-5541** / MIL-DTL-81706) is a *separate process* with its own chemistry, and it's the subject of a different manual. This manual is **chromate on zinc and zinc alloys.** If aluminum parts show up on your line, flag it — they don't belong in a zinc passivate.

• WHERE CHROMATE FITS IN THE SPECS (AND THE FAMILY)

You'll hear job specs called out by standard. Here are the ones that matter for chromate-on-zinc:

SPEC	WHAT IT COVERS	NOTE
ASTM B633	Electrodeposited **zinc on steel**, including supplementary **chromate Types** and thickness **Classes**	The everyday workhorse spec for zinc + chromate
ASTM F1941 / F1941M	Electrodeposited coatings (zinc, etc.) on **threaded fasteners**	When the parts are screws, bolts, nuts
MIL-DTL-5541 (and 81706)	Chromate/chem-film on **aluminum**	**Different metal — separate manual.** Don't confuse with zinc chromate.

On the ASTM B633 "Types": B633 separates the **Class** (the *zinc thickness* / service condition) from the **Type** (the *supplementary finish* on top of the zinc). The two classic chromate Types are **Type II — colored chromate** (the iridescent yellow and olive-drab family) and **Type III — colorless / clear chromate** (the clear/blue-bright family).

> ⚠ **HEADS-UP — VERIFY THE EXACT TYPE NUMBERS**
>
> ASTM revises B633 periodically, and newer editions have **added Types to cover trivalent (Cr III) passivates and to re-organize colored vs. colorless designations.** Do **not** quote a Type number from memory on a job — pull the **current B633 revision** (and the customer's purchase-order callout) and confirm. We're flagging this rather than guessing a number that may be out of date.

> ★ **REMEMBER**
>
> **Class = how thick the zinc is. Type = what passivate/finish goes on top.** A spec like "zinc per ASTM B633, [Class], Type II yellow" is telling you both the zinc thickness *and* that you owe it a colored (yellow) chromate. Read both halves.

• THE HONEST TRADEOFFS

A good operator knows the weaknesses too:

- **It's hexavalent chromium — a carcinogen and a regulatory target.** This is the dominant downside and the reason the whole industry is moving away (Chapter 3 / Part Three).
- **The fresh film is soft and fragile.** It's a hydrated gel when it comes out of the tank; rough handling or hot drying ruins it before it cures (Stations 04–07 / Part Three).
- **Color is not infinitely controllable.** Color tracks film weight and chemistry, so hitting an exact decorative shade run-to-run takes a controlled bath.
- **It's a zinc-family process.** It needs a reactive freshly plated surface; it is *not* a finish for aluminum or bare steel.

> ☀ **FUN FACT**
>
> That shimmering yellow-gold iridescence isn't a dye in the classic yellow chromate — it's a *thin-film interference* effect plus the natural color of trapped Cr(VI), the same physics that paints an oil slick or a soap bubble. You're partly reading the coating's chromium content with your eyes.

• A FIRST LOOK AT THE PROCESS NUMBERS

We'll go deep at Station 04, but here's the shape of the chromate dip so the rest of the manual makes sense:

PARAMETER	TYPICAL RANGE	WHY IT'S LIKE THIS
Bath type	Dilute acidic **hexavalent chromium** (chromic acid / dichromate) + activators	The Cr(VI) + acid + activator trio is what converts zinc
pH	`~1.5–2.5 (strongly acidic)`	Acid drives the zinc-dissolving reaction that starts film growth
Temperature	`Room temp (≈20–30°C / 70–85°F)`	Unlike phosphate/hard-chrome lines, most chromates run cool
Immersion time	`Very short — ~15–60 s`	The reaction is fast; over-immersion powders the film
Drying	`Low temp — below ~60°C (140°F)`	A hot dry **cracks/dehydrates the soft gel** and destroys self-healing

> ★ **REMEMBER**
>
> Three things make a chromate dip different from almost every other tank you'll run: it's **cool, it's quick (seconds, not minutes), and the part must be dried *gently*.** If your instinct from a phosphate or hard-chrome line is "hotter and longer is better," unlearn it here — on a chromate, hot and long *destroy* the coating.

> ⚙ **TECHNICAL STUFF — ACTIVATORS**
>
> Besides the chromium and the acid, chromate baths contain small amounts of **activators / accelerators** (commonly sulfate, plus things like formate, acetate, chloride, nitrate, or fluoride depending on the product). These control how aggressively the acid attacks the zinc and how the film builds — the difference between a bright, tight film and a dull, powdery one. Exact identities and amounts come from the **supplier TDS**; they're maintained by your chemist/lab, not guessed at the tank.

CHAPTER 3 · THE MOST IMPORTANT CHAPTER

WHY HEXAVALENT?

THE CR(VI) HAZARD AND THE ROHS DRIVER, UP FRONT — READ THIS CHAPTER TWICE

We put this chapter near the front on purpose. The defining feature of a classic chromate is that it runs on **hexavalent chromium, Cr(VI)** — and Cr(VI) is a **confirmed human carcinogen.** It is also the single biggest reason this entire chemistry is being **phased out** of commercial industry. You need to understand both of those — the *health* hazard and the *regulatory* hazard — before you read another word about how to run the line.

WHY THE BATH HAS TO BE HEXAVALENT (AND WHY THAT'S THE PROBLEM)

The classic chromate bath gets its power from **hexavalent chromium** — chromium in its **+6 oxidation state** (from chromic acid, CrO_3, and/or dichromate salts). That hexavalent chromium is what makes the film form *and* what gives it self-healing (the trapped Cr^{6+} reservoir from Chapter 2). **The very property that makes the coating work is what makes it dangerous:** mobile, soluble, oxidizing hexavalent chromium.

TECHNICAL STUFF

"Hexavalent" (Cr^{6+}, Cr VI) and "trivalent" (Cr^{3+}, Cr III) describe the chromium atom's oxidation state. **Cr(VI) is a strong oxidizer, highly toxic, and carcinogenic. Cr(III)** — the form chromium ends up in *after* it's been reduced — is far less hazardous and is even a trace nutrient. **The whole goal of waste treatment on this line (Part Three) is to convert the dangerous Cr^{6+} into the safe Cr^{3+} and then remove it.** And the whole goal of the *next* manual's chemistry is to build a passivate out of Cr(III) from the start.

HOW CR(VI) HURTS PEOPLE

This is not abstract. Cr(VI) exposure on a finishing line causes real, documented harm:

- **Lung and sinonasal cancer** — the most serious. Long-term inhalation of chromic-acid mist is the reason it's regulated as a carcinogen.
- **Nasal septum ulceration and perforation** — chromic-acid mist can eat a hole through the cartilage between the nostrils. Once a common, visible mark on long-time chrome workers. **Preventable** with proper controls.
- **"Chrome holes" / chrome ulcers** — deep, slow-healing skin ulcers where chromic acid contacts even a tiny break in the skin.
- **Skin and respiratory irritation, dermatitis, and allergic sensitization.**
- **Eye damage** — chromic acid is corrosive to the eyes.

HEADS-UP — THE HAZARD IS MOSTLY INVISIBLE

Acidic chromate solutions and their **mist/aerosol** are the main route into your lungs, and you can be over-exposed without seeing or strongly smelling anything. **Engineering controls (ventilation + mist suppression) are your primary protection — not just PPE.** If the ventilation or mist control isn't working, the tank is not safe to run. Stop and report it.

THE LAW (PART 1): OSHA'S CR(VI) STANDARD

Cr(VI) has its own dedicated OSHA standard — **29 CFR 1910.1026**. You don't need to memorize the legal text, but you must know the numbers your shop's program is built around:

TERM	VALUE	PLAIN MEANING
PEL (Permissible Exposure Limit)	5 µg/m³	The legal ceiling for Cr(VI) in the air you breathe, averaged over an 8-hr shift
Action Level	2.5 µg/m³	The "pay attention" trigger (8-hr avg) — at/above this, the shop must do air monitoring, training, and medical surveillance

A microgram (µg) is a *millionth* of a gram. A limit of 5 µg per cubic meter of air is *extraordinarily* small — which tells you how potent this hazard is and why mist control is non-negotiable.

⚙ **TECHNICAL STUFF**

Because the PEL is so low, you **cannot** judge your exposure by smell or eye irritation — those happen at far higher levels. That's why OSHA requires *air monitoring*, *medical surveillance*, regulated areas, a written exposure-control plan, and specific hygiene facilities. Your shop's Cr(VI) program is a legal requirement, not a suggestion.

THE LAW (PART 2): THE ROHS / ELV / REACH DRIVER

There's a second body of law that doesn't protect *you* directly but is reshaping the whole industry: **product and environmental rules that restrict hexavalent chromium in finished goods.**

- **RoHS** (Restriction of Hazardous Substances) — restricts Cr(VI) in **electrical and electronic equipment.**
- **ELV** (End-of-Life Vehicles directive) — restricts Cr(VI) in **automotive** parts.
- **REACH** — places hexavalent-chromium substances on the EU **authorization** list, meaning their use requires special, time-limited authorization and is being actively driven down.

The combined effect: **for most commercial markets — electronics, automotive, consumer goods — hexavalent chromate is no longer acceptable.** This is *the* reason the industry developed and largely switched to **trivalent (Cr III) passivates.**

★ **REMEMBER**

There are **two** legal stories on this line, and they point the same direction. **OSHA 1910.1026** protects the *worker* from the carcinogen in the tank. **RoHS / ELV / REACH** restrict the carcinogen in the *product* you ship. Together they're why hex chromate is being phased out — and why our **next manual is the Trivalent (Cr III) Chromate.**

⚠ **HEADS-UP**

Knowing a compliant trivalent alternative *exists* does **not** change how you must treat the hexavalent bath in front of you. The tank you're running is Cr(VI), a carcinogen. Treat it that way, every shift.

• THE FIVE PILLARS OF CR(VI) CONTROL

You'll see these at every station:

1. **Mist suppression at the tank** — a **fume suppressant** (a surfactant additive that lowers surface tension so bursting bubbles throw less mist) and/or floating covers. First line of defense, right at the source.
2. **Tank ventilation** — **local exhaust** (lateral or push-pull hoods) that pulls mist off the surface, often with a **scrubber** that washes Cr(VI) out of the exhaust air before it leaves the building.
3. **Respiratory protection** — the right respirator for the task, per your shop's program, used *in addition to* (never instead of) ventilation.
4. **Hygiene and no hand-to-mouth** — **no eating, drinking, smoking, vaping, or cosmetics** anywhere near the line; designated wash-up; separate work clothing; decontamination before breaks and before leaving.
5. **Spill, waste, and skin protection** — full chemical PPE, immediate flushing of any contact, controlled spill response, and proper **Cr(VI) waste treatment** (reduce $Cr^{6+} \rightarrow Cr^{3+}$, then precipitate — Part Three).

★ **REMEMBER**

Engineering controls first (suppress the mist, ventilate the tank), respirator second, hygiene always. **PPE is the *last* layer of defense, not the first.** A respirator does not make a tank with broken ventilation safe.

• HONEST POSITIONING: HEX VS. TRIVALENT

	HEXAVALENT CHROMATE (THIS MANUAL)	TRIVALENT PASSIVATE (NEXT MANUAL)
Chromium state	**Cr(VI) — carcinogen**	**Cr(III) — far safer**
Self-healing	**Yes — strong** (mobile Cr^{6+})	Weaker; usually **needs a topcoat/seal**
Corrosion-per-cost	**Excellent** — cheap, high protection	Good, but topcoats add cost/steps
RoHS / ELV / REACH	**Restricted**	**Compliant** — the modern standard
Status	**Being phased out**; survives where RoHS doesn't apply (some military/defense, certain industrial)	**The default** for new commercial work

★ **REMEMBER**

Frame it honestly. **Hex chromate is the cheaper, self-healing, best-classic-corrosion-per-dollar option — but it's a carcinogen and RoHS-restricted, so it's being phased out.** Trivalent is the RoHS-compliant modern standard that usually needs a sealer/topcoat to match hex's performance. Hex still has a foothold where RoHS/ELV don't reach (some military and industrial work). Both are legitimate to *know*; only one is the future.

⚠ **HEADS-UP**

If you take only one thing from this chapter: the chromate bath is a **confirmed human carcinogen** whose worst hazard is an *invisible mist*, and it's a **regulated substance** the world is phasing out. Suppress the mist, ventilate the tank, wear your respirator, keep your hands away from your face, and never let Cr(VI) reach a drain. Do those things, every shift, and you can work this line safely.

CHAPTER 4

HOW A CHROMATE LINE WORKS

A LINE IS A SYSTEM, NOT A PILE OF SEPARATE TANKS

A chromate line is a **sequence of stations** a part travels through in a strict order. Because chromate is the *finishing* step, the line usually starts where the **zinc-plating line ends** — the part arrives already plated and rinsed. Parts are typically **barrel-processed** (small hardware/fasteners tumbled in a perforated barrel) or **rack-processed** (larger or cosmetic parts hung individually), the same way they were plated.

> 💡 **TIP**
> The big mental model for this line: **everything before the chromate exists to deliver a chemically *fresh, clean, active* zinc surface; everything after exists to *protect the soft film you just grew until it cures.*** The chromate dip itself takes seconds — most of the operator's care is on either side of it.

THE WHOLE CHROMATE SEQUENCE (TEACHING LAYOUT)

#	STATION	THE ONE THING IT DOES	TEMP	NOTES
01	Receive & Inspect	Confirm the part arrived freshly **zinc-plated & rinsed**, in good condition	Ambient	Time-sensitive — fresh zinc passivates best
02	Nitric Activation / Bright Dip	Quick dilute-acid dip to **desmut/brighten** the zinc and wake up the surface	Ambient	Strong acid; very short
03	Rinse (post-activation)	Wash off the activation acid	Ambient	The firewall before the chromate
04	**Hexavalent Chromate**	**Grow the self-healing chromate film — the passivate**	Amb ~20–30°C	**Main event; Cr(VI); pH 1.5–2.5; ~15–60 s**
05	Rinse (controlled / cool)	Gently rinse the soft film without stripping the Cr^{6+} reservoir	Cool / amb	Over-hot/over-long rinse leaches protection
06	Seal / Topcoat (optional)	Add a sealer/topcoat for extra corrosion life or lubricity	Per product	Chrome or, more often now, non-chrome
07	Dry-Off (low temp)	Drive off water **gently** so the gel cures, not cracks	≤~60°C	**Do NOT over-dry — the key operator rule**
08	Inspect & Hand-Off	Confirm color/coverage and that the film isn't rubbed/powdery	Ambient	Fresh film hardens over ~24 h

The basic rhythm: **Receive → Activate → Rinse → CHROMATE → Rinse → Seal → Dry (cool) → Inspect.**

> ⚠ **HEADS-UP**
> Notice the bath temperature: a chromate runs **cool, around room temperature** — the opposite of a hot phosphate or hard-chrome tank. And notice the **dry-off is deliberately *low temperature.*** Both are easy to get wrong if you're used to other lines. **Cool dip, gentle dry.**

A LINE IS A SYSTEM, NOT A PILE OF SEPARATE TANKS

★ **REMEMBER**

Every station either protects the next or poisons it. There is no neutral step. A part that arrives with stale, oxidized zinc won't passivate evenly. A part that drags activation acid into the chromate shifts the bath. A part rinsed too hot loses its self-healing reservoir. A part dried too hot cracks before it cures. What you do — or fail to do — early shows up as a corrosion failure (or a safety failure) later.

💡 **TIP**

Think of the chromate tank as the crown jewel *and* the dragon of the line. Everything *before* it (receive, activate, rinse) exists to deliver a perfectly fresh, clean, active zinc surface into it. Everything *after* it (rinse, seal, dry, inspect) exists to protect and cure the soft, hazardous film it created. The whole line orbits that one cool, yellow, carcinogenic tank.

WHY "FRESH" ZINC MATTERS

Unlike steel (which you can clean and re-clean), an **electroplated zinc surface ages.** Left sitting, bright zinc starts to oxidize and dull within hours, and a dull, oxidized zinc surface **passivates poorly** — you get thin color, patchy coverage, and weak protection. That's why a chromate line ideally runs **right after** the zinc line, and why the activation dip (Station 02) exists: to strip any light oxide/smut and re-expose fresh, reactive zinc just before the chromate.

★ **REMEMBER**

Fresh zinc passivates well; stale zinc passivates badly. If parts have been sitting (overnight, over a weekend, or came in from another shop), expect to lean harder on the activation step — and flag heavily oxidized or white-rusted parts to your lead, because no chromate fixes badly corroded zinc.

☀ **FUN FACT**

The activation dip is sometimes called a "bright dip" because on a dull, slightly oxidized zinc part you can literally *watch* it flash back to a clean, bright silver in the few seconds it's in the acid. That brightening is the smut and oxide dissolving away, re-exposing the reactive metal the chromate needs.

CHAPTER 5

WHY THE ORDER MATTERS

EACH STATION SETS UP THE NEXT, AND THE SEQUENCE IS LOCKED BY CHEMISTRY — AND BY SAFETY

You might think the station order is just convention. It isn't. Each station exists *because of* the one before and after it.

RECEIVE FIRST — BECAUSE TIME AND CONDITION MATTER

Because fresh, clean zinc passivates well and stale, oxidized, or contaminated zinc passivates badly. You confirm the part is genuinely freshly plated and rinsed, in good condition, *before* any chromate chemistry is wasted on a surface that can't accept it.

ACTIVATE BEFORE CHROMATE — BECAUSE ZINC OXIDIZES

Because even "fresh" zinc carries a light oxide/smut film that blocks an even reaction. The dilute-acid **activation/bright dip** dissolves that film and a whisker of zinc, leaving a chemically fresh, uniformly reactive surface. Skip it, and you get **patchy color and thin protection.**

> ★ **REMEMBER**
>
> **No activation, no uniform passivate.** A clean, freshly activated zinc surface is what lets the chromate form an even, full-color, full-protection film. Activate right before you chromate — an activated surface re-oxidizes in minutes.

RINSE BETWEEN ACTIVATION AND CHROMATE — BECAUSE DRAG-IN SHIFTS THE BATH

Because the activation acid is its own chemistry; carry a film of it into the chromate and you **drift the chromate's pH and contaminate it.** The rinse is the firewall.

> ★ **REMEMBER**
>
> *"Drag activation acid into the chromate and you blunt the chromate."* Rinses keep each chemistry out of the next tank. This is your first firewall on the line.

CHROMATE — THE MAIN EVENT (COOL AND QUICK)

Now — and only now, with a clean, freshly activated zinc surface — the chromate film actually forms. The dip is **cool, strongly acidic, and *short* (seconds).** This is the payoff stage and the carcinogen stage; treat it with total respect.

> ⚠ **HEADS-UP**
> The chromate solution is **hexavalent chromium — a confirmed human carcinogen — and acidic.** Even though it's only seconds of immersion, the mist, the drag-out, and any splash are all Cr(VI). Full PPE, ventilation, and mist suppression every time. (Full detail at Station 04.)

RINSE AFTER CHROMATE — GENTLY, AND COOL

Because the part leaves the chromate wet with Cr(VI) solution that must come off — but the **fresh film is a soft, soluble gel.** A rinse that's too hot, too long, or too aggressive will **leach the soluble hexavalent chromium back out** and weaken both color and self-healing. The rinse must be **cool and controlled** — enough to wash off drag-out, not so much that it strips the reservoir.

> ★ **REMEMBER**
> This rinse is a balancing act unique to chromates: **wash off the carcinogenic drag-out, but don't wash away the protection.** Cool water, short dwell, no hot final rinse. Skimp and you ship Cr(VI) drag-out; overdo it (hot/long) and you strip the self-healing.

SEAL/TOPCOAT (OPTIONAL), THEN DRY COOL, THEN INSPECT

- **Seal/topcoat** if the spec calls for it — extra corrosion life or lubricity.
- **Dry low and slow** so the gel **cures** instead of cracking. (The single biggest operator mistake on this line lives here — Station 07 and Part Three.)
- **Inspect last** to confirm color, coverage, and that the film isn't powdery or rubbed.

> ⚠ **HEADS-UP — THE ORDER *AND* THE TEMPERATURE ARE LOCKED**
> You **dry cool** because the freshly formed film is a **hydrated gel.** Dry it in a hot oven (like you would a phosphated part) and you **dehydrate and craze the film, drive off the soluble Cr^{6+}, and destroy corrosion resistance and self-healing.** On this line, "more heat to dry faster" is a coating-killer. Patience is part of the procedure.

> ⚙ **TECHNICAL STUFF**
> The recurring villain that justifies the rinses is **drag-out** (solution clinging to a part as it leaves a tank) and its mirror, **drag-in** (that film contaminating the next tank). On a Cr(VI) line drag-out is doubly serious: it wastes chromium *and* spreads a carcinogen around the shop and into the wastewater. That's why chromate lines use **dragout-recovery rinses** and careful Cr(VI) waste treatment (Part Three).

CHAPTER 6

PPE — YOUR DAILY ARMOR

THE GEAR THAT STANDS BETWEEN YOU AND A CARCINOGENIC, CORROSIVE ACID

A chromate line works with strong activation acids and — above all — a **carcinogenic hexavalent-chromium bath and its mist.** PPE is your *last* layer of protection (after ventilation and mist suppression), but it is essential.

★ **REMEMBER**

There is no part, no order, and no deadline worth your health. Scrap can be reworked. Your lungs cannot. On this line that's not a slogan — Cr(VI) makes it literal.

THE MASTER PPE SET (WEAR AT ALL CHEMICAL STATIONS)

- **Chemical splash goggles** — sealed goggles, not safety glasses. Glasses leave gaps; chromic acid finds them.
- **Face shield** — *over* the goggles for any work at the chromate, activation, or seal tanks.
- **Chemical-resistant gloves** — elbow-length, rated for chromic/nitric acid (nitrile, neoprene, or PVC per the SDS). Inspect for pinholes before each use — a chrome ulcer starts at a single break.
- **Chemical-resistant apron/suit** — full coverage, chest to below the knee.
- **Chemical-resistant, non-slip boots** — floors around dip tanks and rinses are wet and slick.
- **Dedicated work clothing** — kept separate from street clothes; never worn home. Keeps Cr(VI) out of your car and house.
- **Respiratory protection** — the respirator your shop's Cr(VI) program specifies for the task. Required whenever you could be exposed to chromic-acid mist above the limits, and any time ventilation is degraded.

⚠ **HEADS-UP — RESPIRATOR IS A BACKUP, NOT A LICENSE**

A respirator is required, but it does **not** replace tank ventilation and mist suppression. If the fume suppressant has died (foam gone, mist visible) or the exhaust isn't pulling, the tank is unsafe even with a respirator on. Stop and report it. Engineering controls are primary; the respirator is the last line.

💡 **TIP**

Put gear on *in order*: boots and apron first, then gloves, then goggles, then face shield and respirator last. Take it off in reverse, and rinse your gloves *before* removing them. This keeps contaminated gloves away from your face — critical with a carcinogen.

STATION	PRIMARY HAZARD	WHY IT'S DANGEROUS
01 Receive/Inspect	Mechanical; sharp/heavy parts	Cuts, pinches
02 Nitric Activation	**Strong acid**	Chemical burns; mist irritation
03 Rinse	Residual acid; wet floor	Mild irritant; **slips**
04 Hexavalent Chromate	**Cr(VI) chromic acid + mist; acidic**	**Carcinogen; severe burns; respiratory harm**
05 Rinse	**Cr(VI)-contaminated water**	Carcinogen; burns; slips
06 Seal/Topcoat	Acidic seal; **Cr(VI) if chromate seal**	Burns; Cr(VI) carcinogen
07 Dry-Off	Warm/hot air, hot parts	Burns; heat stress
08 Inspect/Hand-Off	Residual chemistry on parts	Mild irritant

⚠ **HEADS-UP**

Never eat, drink, smoke, vape, or apply cosmetics on the line. Hand-to-mouth contact is a major route for ingesting Cr(VI). Wash thoroughly before breaks and before leaving. Keep all food and drink completely out of the work area, and keep work clothes separate from street clothes.

CHAPTER 7

EMERGENCY & FIRST-RESPONSE

WHEN SECONDS COUNT — LEARN THIS COLD, BEFORE YOU EVER NEED IT

Know these *before* you need them. **Locate, before your first shift:** the nearest **emergency eyewash** (within ~10 seconds of every chemical station), the **safety shower**, the **fire extinguisher** and alarm, the **spill kit**, the **SDS binder**, and your shop's **Cr(VI) regulated-area** boundaries and decon station.

⚠ **HEADS-UP**

Eyewash and shower stations must always be unblocked. Never stack baskets, totes, or boxes in front of them. The day you need one is the day you can't move boxes out of the way.

• FIRST RESPONSE BY HAZARD

SITUATION	WHAT TO DO — IMMEDIATELY
Chromate / acid on skin	Safety shower; flush **at least 15 minutes.** Remove contaminated clothing while flushing. Even small chromic-acid contact on broken skin needs medical attention (chrome-ulcer risk). Bring the SDS.
Chromate / acid in eyes	Eyewash; hold eyelids open and flush **at least 15 minutes**, rolling the eyes. Do not rub. Get medical attention immediately — chromic acid is corrosive to eyes. Bring the SDS.
Inhaled chromic-acid mist, feel ill	Move to fresh air immediately. Report it — do not "tough it out." Seek medical attention for anything beyond brief mild irritation; report as a possible Cr(VI) exposure.
Cr(VI) swallowed	**Do not** induce vomiting unless the SDS directs it. Follow SDS first aid; call poison control / medical help immediately.
Nosebleed / sore nasal passages	Report it — recurrent nosebleeds or nasal soreness can be an early sign of Cr(VI) over-exposure. It triggers a ventilation/exposure check. Don't ignore it.
Chemical spill	Alert others; keep people away. Clean only if trained and within your shop's "small spill" limits — otherwise evacuate and call trained spill response. **Never let chromate spills reach a floor drain.**

⚠ **HEADS-UP — LOCKOUT/TAGOUT (LOTO)**

Mandatory for *all* electrical/mechanical maintenance — heaters, pumps, hoists, dryer, ventilation. Physically lock the power off and tag it so no one can re-energize while someone works. "I'll just be a second" is how people get hurt.

⚠ **HEADS-UP — A CRITICAL ACID-MIXING RULE, FOR LIFE**

When diluting acid or making bath additions per your procedure, **always add ACID/chemical to WATER — never water to concentrated acid.** Adding water to concentrated acid releases heat so violently it can erupt acid into your face. Memory aid: *"Do as you oughta — add acid to water."*

☀ **FUN FACT**

The 15-minute flush rule isn't arbitrary — it's the time emergency-response standards found necessary to dilute and carry away most corrosive chemistry before it does deep tissue damage. That's why eyewash and shower stations must deliver a steady, hands-free flow for a full 15 minutes.

CHAPTER 8

THE SAFETY PHILOSOPHY OF A CR(VI) CHROMATE LINE

A QUICK DIP IS STILL A CARCINOGEN — TREAT IT THAT WAY

People sometimes assume that because the chromate dip is fast, cool, and “just a rinse-like step,” it’s the safe one. That’s a dangerous assumption. **The dip is short, but the chemistry is the most hazardous in the whole conversion-coating family.**

- **The bath is a carcinogen.** Hexavalent chromium is a confirmed human carcinogen whose worst hazard — the mist — is invisible. Ventilation and mist suppression are primary controls, not afterthoughts.
- **The acids bite.** The activation/bright dip and the chromate itself are strongly acidic — burns and eye injuries are real.
- **Drag-out spreads the hazard.** Every part carries a film of Cr(VI) out of the tank. Sloppy handling spreads a carcinogen across the shop, onto floors, and into the rinses.
- **The waste is regulated.** Cr(VI) must be **reduced to Cr(III) and precipitated** before discharge; it can never go down a drain. (Part Three.)
- **The product is regulated.** RoHS/ELV/REACH mean what you make may be restricted — know whether your shop runs hex for a market that still allows it.

★ **REMEMBER**

The safety philosophy of this line is simple: **respect the carcinogen, control the mist, contain the drag-out, treat the waste, and keep your hands away from your face.** A chromate line isn’t “the quick easy one” — it’s a *Cr(VI)* line that happens to be quick, and it deserves the same discipline you’d give a tank full of any carcinogen.

★ **REMEMBER**

You’ve now got the whole foundation: what a conversion coating is, what makes a chromate special (self-healing via trapped Cr^{6+}), how the eight-station line works as one connected system, why the order is locked by chemistry, the Cr(VI) hazard and the RoHS driver, what to wear, what to do in an emergency, and the safety mindset that ties it together. Turn the page knowing the *why* — and the *how* will make a lot more sense.

PART TWO — THE LINE, STATION BY STATION (STATIONS 01–08)

STATION 01 OF 08

RECEIVE & INSPECT

"YOU CAN'T PASSIVATE ZINC THAT'S ALREADY GONE BAD."

Before any chromate chemistry touches a part, someone confirms it arrived **freshly zinc-plated and rinsed**, in the right condition, and on time. On most lines the chromate process picks up directly where the zinc line dropped off. This feels like the un-glamorous step. It isn't: a chromate is only as good as the zinc surface it's given, and **zinc ages.**

• WHAT YOU'RE DOING & WHY IT MATTERS

- **Confirm the job.** Right part number, right quantity, right spec. Check the traveler for the **target color/Type** (clear, yellow, olive, black), the **substrate** (zinc? zinc-nickel? cadmium?), and any seal/topcoat requirement.
- **Confirm freshness.** Ideally the parts come straight from plating and rinse. Parts that have sat (overnight, a weekend) or that show **dullness or white rust** passivate poorly — flag them.
- **Inspect condition.** White rust, heavy water spotting, fingerprints, or contamination on the zinc are warning signs. Heavily corroded zinc cannot be rescued by a chromate.

• OPERATOR ESSENTIALS

ITEM	GUIDANCE
Substrate check	Zinc / zinc-alloy / cadmium = good. Aluminum or bare steel = **wrong line — flag it**
Freshness	Straight from plating is best; aged/dull zinc → lean on activation, flag if white-rusted
Target color/Type	Confirm clear / yellow / olive / black per traveler before you start
Handling	Clean gloves — fingerprints on zinc become defects under the chromate
Load/basket condition	Parts not nested; barrels not overpacked so solution reaches all faces

⚠ **HEADS-UP**

Mechanical hazards rule this station — sharp edges, heavy baskets, pinch points. Wear cut-resistant gloves for handling, lift with your legs, and get help for heavy loads.

• STEP-BY-STEP PROCEDURE

1. **Read the traveler.** Confirm part, quantity, substrate, target color/Type, and any seal/topcoat callout.
2. **Check freshness and condition.** Bright, recently plated zinc is ideal. Set aside anything dull, white-rusted, or contaminated for your lead.
3. **Handle with clean gloves.** Avoid bare-hand contact — skin oils cause spotty passivation.
4. **Load for full contact and drainage.** No nesting; barrels not overpacked; every surface must see solution and every cavity must drain.
5. **Flag anything unusual** (wrong substrate, aged parts, mixed lots) before the load enters the activation dip.

• TOP MISTAKES NEW OPERATORS MAKE

1. **Running aged or white-rusted zinc** as if it were fresh — thin color, patchy protection.
2. **Bare-handed handling** — fingerprints become passivation voids.
3. **Loading the wrong substrate** (aluminum/steel) without flagging it.
4. **Overpacking barrels / nesting parts** so solution can't reach all faces.
5. **Letting fresh parts sit** before processing — the zinc oxidizes while they wait.

• TEACH-BACK CHECKLIST

- [ ] Explain why **fresh** zinc passivates better than aged zinc.
- [ ] State which substrates belong on this line (zinc/zinc-alloy/cadmium) and what to do with aluminum or steel (flag it).
- [ ] Identify what incoming conditions need flagging (dullness, white rust, contamination).
- [ ] Describe why clean-glove handling matters.

SIGN-OFF — STATION 01

Trainee: ______________________ Trainer: ______________________ Date: ____________

"I confirm I verified the job, substrate, target color, and that parts were fresh and in good condition, handled cleanly, and loaded for full contact and drainage."

STATION 02 OF 08

NITRIC ACTIVATION / BRIGHT DIP

“WAKE THE ZINC UP — STRIP THE SMUT, EXPOSE FRESH METAL.”

This is a quick dip in a **dilute acid** (commonly a few percent **nitric acid**, per your TDS) that **dissolves the light oxide/smut film** on the zinc and a whisker of zinc itself, leaving a bright, chemically fresh, uniformly reactive surface for the chromate. On some lines it’s also called a “bright dip.” It is short — usually just a **few seconds.**

• WHAT YOU’RE DOING & WHY IT MATTERS

Even fresh zinc carries an invisible oxide/smut layer that makes the chromate form unevenly — patchy color, thin spots, weak protection. The activation dip removes it and **resets the surface** so the chromate reacts the same way everywhere.

TECHNICAL STUFF

A dilute nitric dip lightly **etches and brightens** zinc, removing oxide/smut and any dragged-in alkalinity from upstream rinses. It also slightly *lowers* the surface so the chromate’s own acid attack starts from clean metal. Concentration is low and time is short on purpose — the goal is to *clean and activate*, not to strip meaningful zinc thickness. Too strong or too long and you remove plated zinc you were paid to apply. **Exact concentration and time come from the supplier TDS.**

• OPERATOR ESSENTIALS

PARAMETER	RANGE	NOTES
Acid	`Dilute nitric (per TDS), ~0.5–3%`	Some products use other dilute acids — follow TDS
Temperature	`Ambient`	No heat needed
Time	`A few seconds (per TDS)`	Short — over-dipping removes zinc
Movement	`Brief agitation/rotation`	Even contact on all faces

HEADS-UP — STRONG ACID

The activation dip is **acidic** and can cause **chemical burns** and serious eye injury, and it throws a fine mist. **PPE every time:** sealed goggles, face shield, chemical gloves, apron, boots. Know your eyewash and shower locations. **Nitric acid is also an oxidizer** — keep it away from organics/combustibles and never mix chemistries.

STEP-BY-STEP PROCEDURE

1. **Check the dip** — concentration in spec (last titration), level, and cleanliness.
2. **Confirm the prior rinse** delivered a clean part — drag-in of alkaline rinse weakens the dip.
3. **Dip for the specified short time** with brief agitation. Watch dull parts flash bright.
4. **Withdraw and drain** back to the tank.
5. **Move promptly to rinse** — an activated zinc surface re-oxidizes within minutes.

TOP MISTAKES NEW OPERATORS MAKE

1. **Over-dipping** — removes plated zinc and can leave the surface over-etched.
2. **Letting activated parts sit** before chromating — they re-oxidize and passivate unevenly.
3. **Running the dip out of spec** (too weak = no activation; too strong = over-etch).
4. **Skipping PPE** because “it’s just a quick dip” — it’s still strong acid.
5. **Dragging alkaline rinse water in** and neutralizing the dip.

QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Dull/smutty parts after dip	Dip too weak or spent	Titrate; refresh per TDS
Over-etched / hazy zinc	Dip too strong or time too long	Reduce time/concentration per TDS
Patchy chromate downstream	Uneven activation; parts sat too long	Re-activate; shorten transfer time

SIGN-OFF — STATION 02

Trainee: ____________________ Trainer: ____________________ Conc./Time: __________

“I confirm the activation dip was in concentration, parts were dipped the correct short time and moved promptly to rinse, and I wore required PPE.”

STATION 03 OF 08

RINSE (POST-ACTIVATION)

“A RINSE ISN’T A REST STOP. IT’S A FIREWALL.”

You just activated the zinc — but it’s coated in a film of activation acid. This station’s job is to **wash that acid off** before the part hits the chromate, so the activation chemistry doesn’t drift the chromate bath. It looks like a “nothing” step. It isn’t: drag-in of acid quietly shifts the chromate’s pH out of its narrow window.

• WHAT YOU’RE DOING & WHY IT MATTERS

The activation acid clinging to the part is **drag-out** (leaving the dip) / **drag-in** (arriving at the chromate). The rinse knocks it down by dilution: clean water surrounds the part, the acid film dissolves into the rinse, and flow flushes it away — *but quickly*, because activated zinc re-oxidizes if it lingers.

💡 **TIP — COUNTER-FLOW RINSING**

The professional approach is a **counter-flow (counter-current) rinse**: parts move forward, water flows backward. Fresh water enters the last (cleanest) stage and overflows into the dirtier one, so the cleanest water the part sees is the last thing to touch it before the chromate. Saves enormous water and gives the chromate a clean start.

```
FROM         +-------------+  PARTS ->  +-------------+         TO
ACTIVATE ->  |  RINSE 1    | ---------> |  RINSE 2    | ---->  CHROMATE
             |  (dirty)    |           |  (clean)    |
             |  acid in    |           |  near-neutral|
             +-------------+           +-------------+
                  ^   overflow <----------    ^
                  +------ <- FRESH WATER FLOWS BACKWARD <- -----+
                                              FRESH IN
```

⚠ **HEADS-UP**

Rinse water carries dragged-in acid and is mildly acidic. Wear **goggles, gloves, and non-slip footwear** — the floor around rinses is always wet, and **slips are the #1 injury here.**

★ **REMEMBER**

Don’t let the part dwell in the rinse. Activated zinc is reactive and re-oxidizes fast. Rinse clean, then move straight to the chromate. A long pause here undoes the activation you just did at Station 02.

• TOP MISTAKES

1. Letting the activated part sit in or after the rinse (re-oxidation).
2. A tired, contaminated rinse that drags acid into the chromate.
3. Treating the rinse as optional “downtime.”

SIGN-OFF — STATION 03

Trainee: ____________________ Trainer: ____________________ Date: ____________

“I confirm the rinse was clean/flowing, parts were rinsed and moved promptly to the chromate, and I wore required PPE.”

STATION 04 OF 08 · THE MAIN EVENT

HEXAVALENT CHROMATE — THE PASSIVATE

COOL, QUICK, CARCINOGENIC — THE SELF-HEALING FILM FORMS HERE

This is the heart of the line and the most hazardous tank in the conversion-coating family. With a clean, freshly activated zinc surface, you dip the part in a **dilute acidic hexavalent-chromium solution** for **seconds**, and the self-healing chromate film grows. Then you treat the part — and yourself — with total respect on the way out.

• WHAT YOU'RE DOING & WHY IT MATTERS

The chromate solution (**chromic acid and/or dichromate + activators, pH ~1.5–2.5, room temperature**) reacts with the zinc: acid dissolves a little zinc, local pH rises, some Cr^{6+} reduces to Cr^{3+}, and a **gel-like chromium/zinc chromate film precipitates — trapping soluble Cr^{6+} that gives the film its self-healing.** Color (clear → yellow → olive → black) tracks film weight, chromium content, and any additives.

• OPERATOR ESSENTIALS

PARAMETER	TYPICAL RANGE	NOTES
Chemistry	Dilute **Cr(VI)** (chromic acid / dichromate) + activators	Per supplier TDS
pH	~1.5–2.5	The master control variable; tightly held
Temperature	Ambient (~20–30°C / 70–85°F)	Cool — *not* a hot tank
Immersion time	~15–60 s (color/Type dependent)	Short; over-immersion powders the film
Agitation	Gentle (per equipment)	Even contact; barrels rotate slowly
Color/Type	Clear / yellow / olive / black per job	Heavier color = more time/chemistry, more protection

⚠ **HEADS-UP — THE SINGLE MOST DANGEROUS TANK ON THIS LINE**

This bath is **hexavalent chromium, a confirmed human carcinogen**, and **acidic.** The hazard is mostly the **invisible mist/aerosol. Engineering controls first:** mist suppressant on the bath surface and local exhaust ventilation must be working *before* you run it. **Then** PPE: sealed goggles, face shield, elbow-length chromic-acid-rated gloves, apron, boots, and the respirator your Cr(VI) program specifies. If mist is visible or exhaust is weak, **stop and report it.**

• STEP-BY-STEP PROCEDURE

1. **Verify controls first.** Confirm ventilation/exhaust is pulling and the **mist suppressant** is present on the bath. No controls = no run.
2. **Check the bath.** pH in spec (last reading), temperature ambient, level and clarity normal.
3. **Confirm the part is freshly activated and rinsed** — and not sitting/re-oxidized.
4. **Immerse for the specified short time** for the target color/Type, with gentle agitation. Watch the color develop.
5. **Withdraw smoothly and let drag-out drain back into the tank** — recovering chromium and keeping the carcinogen contained.
6. **Move directly to the controlled rinse.** Do not let the soft film dry on the way (it'll spot) and do not let drag-out drip across the floor.

💡 TIP — READ THE COLOR AS IT FORMS

On a yellow chromate you can watch the part go from bright silver to pale straw to full iridescent gold in the few seconds it's in the tank. **Pull at the target color.** Under-color = too short/too weak (thin protection); chalky/powdery = too long/too aggressive (over-attacked film). The color is your real-time gauge.

⚠ HEADS-UP — THE FRESH FILM IS SOFT

What you just grew is a **hydrated gel**, not a hard coating. It is easily **wiped, scratched, or marked** until it cures. Handle gently from here to dry-off; don't stack, rub, or let parts bang together. It hardens over the next several hours to ~24 hours.

⚙ TECHNICAL STUFF — WHY SHORT AND COOL

The film-forming reaction is fast at this acidity. **Over-immersion keeps dissolving zinc faster than the film can stabilize**, giving a loose, powdery, low-adhesion film and over-thinning the zinc. Running cool keeps the attack controlled and the film tight. This is the opposite of a phosphate or hard-chrome tank, where heat and time build the coating — here they *destroy* it.

• DIALING THE COLOR / TYPE

TARGET COLOR	ROUGHLY HOW YOU GET THERE	RELATIVE PROTECTION
Clear / blue	Shortest dip / lighter chemistry (sometimes a bleach/leach step removes yellow)	Lowest
Yellow / gold iridescent	Standard dip time, full chemistry	High
Olive-drab / bronze	Longer dip / heavier chemistry / specific additives	Highest
Black	Special additive chemistry (e.g. silver salts) over the chromate	Good (spec-dependent)

★ **REMEMBER**

Heavier and darker = more chromium = more protection; clear = least. When a job calls for more salt-spray life, it usually calls for a *more colored* chromate (or a seal/topcoat). When it calls for a bright, decorative look, expect a clear/blue with lower bare protection. Match the color to the spec — don't just run "whatever the tank gives."

⚙ **TECHNICAL STUFF — THE CLEAR "BLUE BRIGHT" ROUTE**

Some clear/blue finishes are made by forming a normal (yellowish) chromate and then briefly **leaching/bleaching** out the yellow Cr(VI) color. That improves appearance but **removes some of the self-healing reservoir**, which is part of why clear chromates protect less than yellow. It's a direct, visible illustration that *the yellow color is the protection*.

• DRAG-OUT: THE CARCINOGEN LEAVING THE TANK

Every part carries a film of Cr(VI) solution out of the chromate. That **drag-out** is wasted chromium, a contamination source, and a carcinogen.

- **Let parts drain over the tank** before transferring.
- **Use the dragout-recovery rinse** if your line has one (it returns chromium to the bath).
- **Never let drag-out drip across the floor or toward a drain.**

⚠ **HEADS-UP**

Cr(VI) drag-out that reaches a floor drain is a **reportable environmental release** and a worker-exposure hazard. Drain over the tank, use recovery rinses, and report any spill. The chromium must end up either back in the tank or in the **waste-treatment system** (reduce → precipitate, Part Three) — never in the sewer.

• TOP MISTAKES NEW OPERATORS MAKE

1. **Running with weak/absent ventilation or mist suppressant** — the cardinal Cr(VI) sin.
2. **Over-immersing** — powdery, low-protection film and thinned zinc.
3. **Letting the soft film dry or get rubbed** before the controlled rinse.
4. **Sloppy drag-out** — spreading the carcinogen and wasting chromium.
5. **Ignoring color** — shipping under-colored (under-protected) parts.

• QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
No/weak color	Dip too short/weak; cold; stale or poorly activated zinc	Lengthen to target color; check pH/chemistry; re-activate
Powdery / rub-off film	Over-immersion; bath too aggressive; pH off	Shorten dip; verify pH per TDS; check temperature
Patchy / uneven color	Poor activation; nesting; uneven agitation	Re-activate; reload; check agitation
Visible mist over tank	Mist suppressant depleted; exhaust weak	**Stop. Restore mist suppressant/ventilation, then resume**

• TEACH-BACK CHECKLIST

- [ ] State that the bath is **hexavalent chromium, a confirmed human carcinogen**, and name the **primary** controls (ventilation + mist suppression) *before* PPE.
- [ ] Explain why the dip is **cool and short**, and what over-immersion does.
- [ ] Explain how **color relates to protection** and how to pull at target color.
- [ ] Describe how **self-healing** works (trapped Cr^{6+} leaches to scratches) and why over-rinsing/over-drying harms it.
- [ ] Describe correct **drag-out** handling and why Cr(VI) can never reach a drain.

⚠ **HEADS-UP — SAFETY GATE**

No operator runs Station 04 solo until the **Cr(VI) program / PPE / ventilation** row of the Training Matrix is signed. This is the carcinogen station — safety qualification comes first, always.

SIGN-OFF — STATION 04

Trainee: ____________________ Trainer: ____________________ pH/Temp: __________ Color: __________

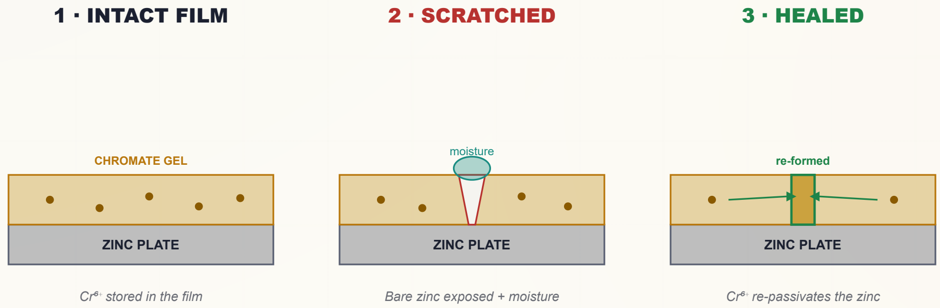
"I confirm ventilation and mist suppression were working, the bath was in pH/temperature spec, I dipped to the target color for the correct short time, handled drag-out correctly, and wore full Cr(VI) PPE."

STATION 04 · THE SIGNATURE PROPERTY

HOW THE FILM PATCHES A SCRATCH

ACTIVE CORROSION PROTECTION — THE COATING THAT FIGHTS BACK

SELF-HEALING — HOW A CHROMATE PATCHES A SCRATCH



Active protection — the coating fights back. This is what trivalent passivates struggle to fully match.

★ **REMEMBER**

This one picture is the whole reason hexavalent chromate survived for a century: **the film carries its own repair kit.** It's also the whole reason it's so hazardous and so regulated — that mobile Cr^{6+} doing the healing is the carcinogen. The property and the problem are the same molecule.

FUN FACT

Because the healing depends on the Cr^{6+} being mobile in moisture, a hex-chromate part can keep re-protecting minor scratches over its service life — a kind of slow-release first-aid for corrosion. Engineers spent decades trying to reproduce that "active" behavior in safer chemistries; the closest modern answers are trivalent passivates *plus* sealers and topcoats (Part Three).

STATION 05 OF 08

RINSE (CONTROLLED / COOL)

"WASH OFF THE CARCINOGEN — WITHOUT WASHING OFF THE PROTECTION."

This rinse is a balancing act found only on chromate lines. You must **remove the Cr(VI) drag-out** clinging to the part — but the fresh film is a **soft, soluble gel holding the self-healing reservoir.** A rinse that's too hot, too long, or too aggressive will **leach the soluble hexavalent chromium back out**, weakening color and self-healing. So: **cool water, short dwell, gentle flow.**

WHAT YOU'RE DOING & WHY IT MATTERS

- **Remove drag-out** so you don't ship Cr(VI)-wet parts or contaminate downstream.
- **Preserve the film** — keep the rinse cool and brief so the Cr^{6+} reservoir stays in the coating.
- **Set up drying** — a clean, lightly rinsed surface dries spot-free.

> ⚠ **HEADS-UP — NEVER A HOT FINAL RINSE ON A FRESH CHROMATE**
> A hot rinse does the same damage a hot oven does: it **leaches the soluble chromium and degrades the film.** Some operators are tempted to use warm water "to help it dry." Don't. The dry-off oven (set *low*) does the drying — the rinse stays cool.

> ★ **REMEMBER**
> **Cool and brief.** Enough to wash off the carcinogenic drag-out, not so much that you strip the self-healing. If you over-rinse a yellow chromate you can literally watch the color fade — that fade is your protection leaving.

> ⚠ **HEADS-UP**
> This rinse water is **Cr(VI)-contaminated.** It goes to **waste treatment** (reduce → precipitate), never down a drain. Wear goggles, gloves, and non-slip footwear; the floor is wet and the water is a carcinogen.

TOP MISTAKES

1. **Hot or prolonged rinse** — fades color, strips self-healing.
2. **Skimping** — leaves Cr(VI) drag-out on the part.
3. **Rough handling** of the still-soft film.

SIGN-OFF — STATION 05

Trainee: ______________________ Trainer: ______________________ Date: ______________

"I confirm I rinsed cool and briefly to remove drag-out without stripping the film, handled the soft coating gently, and routed rinse water to waste treatment."

STATION 06 OF 08

SEAL / TOPCOAT (OPTIONAL)

“WHEN THE CHROMATE ALONE ISN’T ENOUGH — OR WHEN YOU WANT LUBRICITY OR EXTRA LIFE.”

Some jobs call for a **seal or topcoat** over the chromate to add corrosion life, control coefficient of friction (important on fasteners), or improve appearance. Whether you run one — and which one — comes from the **job spec and TDS.**

• WHAT YOU’RE DOING & WHY IT MATTERS

A seal/topcoat is a thin additional layer applied over the rinsed chromate that **closes pores and adds a barrier** on top of the self-healing film. On hex chromate it’s optional (the hex film already performs well); on **trivalent** passivates a seal/topcoat is often **required** to reach equivalent performance — a key point for the next manual.

SEAL/TOPCOAT TYPE	WHAT IT ADDS	NOTE
Non-chrome organic / polymer topcoats	Corrosion life, color, lubricity (torque-tension control)	Increasingly the default
Silicate / inorganic seals	Barrier, corrosion life	Common, chrome-free
Chromic / chromate seals (Cr VI)	Extra protection	**Carcinogen — same Cr(VI) hazards/controls as the main tank**

⚠ **HEADS-UP**

If your seal is a **chromic/chromate (hexavalent, Cr VI)** product, you are handling a **confirmed human carcinogen** with the *same* OSHA controls as Station 04. Many shops run **non-chrome** seals specifically to limit Cr(VI) exposure to one tank. **Know which seal your shop runs.**

★ **REMEMBER**

On **hex** chromate the seal is usually **optional polish**; on **trivalent** it’s often **mandatory** to hit the spec. That difference is one more reason the hex film “does more on its own” — and one more cost/step the trivalent route trades for RoHS compliance.

• OPERATOR ESSENTIALS

PARAMETER	GUIDANCE
Run it at all?	Only if the **spec/traveler** calls for it
Type	Per TDS — non-chrome unless spec requires chromate seal
Temperature/time	Per product (some are warm dips, some dried-in-place)
Cr(VI)?	If a chromate seal: full Cr(VI) controls and PPE

• STEP-BY-STEP PROCEDURE

1. **Check the traveler** — is a seal/topcoat required, and which?
2. **Confirm the chromate film is intact** and properly rinsed (cool) first.
3. **Apply per TDS** — correct concentration, temperature, dwell.
4. **Drain** and move to dry-off. Many topcoats are **dried-in-place** (no rinse after).
5. **Handle gently** — the stack (chromate + seal) is still curing.

• TOP MISTAKES

1. **Running a seal the spec didn’t call for** (or skipping one it did).
2. **Treating a chromate seal as harmless** — it’s still Cr(VI).
3. **Sealing over a damaged/rubbed chromate** — you lock in the defect.

• TEACH-BACK CHECKLIST

- [ ] Explain when a seal/topcoat is run (only when the spec calls for it).
- [ ] Name the seal families and which avoid Cr(VI) (non-chrome organic/polymer, silicate).
- [ ] State the special hazard/controls for a chromate (Cr VI) seal.
- [ ] Explain why hex usually treats the seal as optional but trivalent often requires it.

SIGN-OFF — STATION 06

Trainee: ______________________ Trainer: ______________________ Seal type: __________

“I confirm I applied the seal/topcoat only as the spec required, per TDS, with the correct Cr(VI) controls if a chromate seal, over an intact chromate film.”

STATION 07 OF 08 · THE MAKE-OR-BREAK STEP

DRY-OFF — LOW TEMPERATURE

"DRY IT GENTLY, OR YOU'LL UNDO EVERYTHING." — THE MOST IMPORTANT OPERATOR LESSON ON THIS LINE

This is where new operators ruin good parts. The freshly formed chromate is a **soft, hydrated gel.** It must be dried **gently and at low temperature** so it **cures** (slowly dehydrates and hardens) instead of **cracking.** Get this wrong and the corrosion resistance and self-healing you grew at Station 04 are destroyed — invisibly.

• WHAT YOU'RE DOING & WHY IT MATTERS

Drying removes surface water so the film can cure and the part can be handled, packed, and (if needed) painted. The catch: **too much heat dehydrates the gel too fast, cracking and crazing the film, driving off the soluble Cr^{6+}, and collapsing the self-healing.** The part may *look* fine and **fail salt spray.**

PARAMETER	GUIDANCE
Temperature	**Low — typically below ~60°C (140°F)**; follow TDS (some products specify even lower)
Method	Warm-air dry, centrifugal spin-dry, or ambient — gentle
Time	Long enough to dry, not so hot it bakes
What NOT to do	**No hot bake.** No phosphate-style 250–400°F oven on a fresh chromate

⚠ **HEADS-UP — THE RULE OF THIS LINE**

Do not over-dry / over-bake a fresh chromate. Over-drying **dehydrates and cracks the soft gel film and destroys the self-healing and corrosion resistance.** Keep the dry-off **below your product's TDS limit (commonly ~60°C / 140°F).** If your instinct from a phosphate or paint line says "crank the oven," ignore it here.

⚙ **TECHNICAL STUFF**

A fresh chromate is a hydrated gel of chromium/zinc oxides and hydroxides holding water and mobile Cr^{6+}. Cured slowly, it dehydrates into a coherent, adherent film. **Flash-dehydrated by high heat, it shrinks, crazes (microcracks), and loses the bound water and soluble chromium that make it protective and self-healing.** The damage is often invisible — which is why the temperature limit is a *hard rule*, not a guideline.

★ **REMEMBER**

Fresh film is soft; it hardens on aging. Even after a proper low-temp dry, the coating keeps curing and hardening over the next several hours to ~24 hours. Handle dried parts gently and avoid heavy stacking/abrasion for the first day. Don't pack hot, wet, or freshly dipped parts tightly.

SIGN-OFF — STATION 07

Trainee: ____________________ Trainer: ____________________ Dryer temp: __________

"I confirm I dried below the TDS temperature limit, did not over-bake the film, and handled the soft, curing coating gently."

STATION 08 OF 08 · THE FINISH LINE

INSPECT & HAND-OFF

"LAST CHANCE TO CATCH IT BEFORE IT SHIPS."

A final check confirms the chromate is the **right color/Type**, **uniform and fully covering**, and **not powdery or rubbed**, then the parts are packed or sent on (to paint, assembly, or shipping).

• WHAT YOU'RE CHECKING

CHECK	WHAT IT TELLS YOU	HOW
Color / Type	Right finish and roughly right protection level	Visual vs. standard (clear/yellow/olive/black)
Coverage / uniformity	No bare or thin spots	Visual, all faces
Rub / powder test	Film cured and adherent, not over-dipped/over-dried	Light wipe — color should **not** rub off as powder
No white rust	Zinc still protected	Visual
Salt spray (per spec)	Actual corrosion life	ASTM B117 on samples (lab)

💡 **TIP**

The fast shop checks are **color** (is it the right shade and full?) and a **light rub** (does color come off as powder?). A correct, uniform color that doesn't rub off says the chromate formed and cured. **Powdery rub-off** means over-immersion or over-drying — pull the lot and look upstream.

★ **REMEMBER**

Remember the film is **still curing.** Don't judge final hardness the instant it leaves the dryer, and don't pack tightly while parts are warm/fresh. Give the coating its cure time before rough handling.

⚠ **HEADS-UP**

Even finished, the parts carry trace chromium. Maintain hygiene — **no eating/drinking at inspection**, wash hands, and follow your shop's handling rules for chromated parts.

• TEACH-BACK CHECKLIST

- [ ] Identify the four classic colors and what each implies for protection.
- [ ] Demonstrate the rub/powder check and say what powder means.
- [ ] Explain why fresh film keeps curing and how that affects handling/packing.

SIGN-OFF — STATION 08

Trainee: ______________________ Trainer: ______________________ Date: ____________

"I confirm I verified color/Type, full coverage, and a non-powdery cured film, checked for white rust, and handled/packed the curing parts correctly."

PART THREE — ACROSS THE LINE
READING AND CONTROLLING THE FILM

COLOR & COATING CONTROL

COLOR IS YOUR GAUGE — IT TELLS YOU FILM WEIGHT, CHROMIUM CONTENT, AND PROTECTION

Because the film is so thin, you "read" a chromate mostly by **color**, backed by lab checks. Color is set by **film thickness + trapped Cr(VI) + additives/dyes.**

COLOR / TYPE	RELATIVE FILM WEIGHT	RELATIVE PROTECTION	TYPICAL SALT-SPRAY TO WHITE RUST (APPROX., SPEC-DEPENDENT)
Clear / blue (B633 Type III, colorless)	Lightest	Lowest	`~12–24 hr`
Yellow / gold iridescent (Type II, colored)	Medium	High	`~96–150+ hr`
Olive-drab / bronze (Type II, colored)	Heaviest	Highest	`~150–300+ hr`
Black	Medium–heavy	Good (spec-dependent)	`varies`

⚠ HEADS-UP — VERIFY THE NUMBERS

The salt-spray hours above are **classic ballpark ranges for illustration only.** Actual performance depends on zinc thickness (the B633 Class), the specific product, sealing/topcoat, drying, and the test detail. **Always qualify to the customer's spec and your own salt-spray data — never quote these numbers as a guarantee.**

★ REMEMBER

Color tracks protection. Clear = least; yellow = high; olive = most; black is its own case. A washed-out or faded color usually means a thin film or one that's been over-rinsed/over-dried — i.e., under-protected — even if it "looks finished." Match color to the spec'd Type.

⚙ TECHNICAL STUFF

Levers that move color/weight: **dip time, bath concentration, pH, temperature, and zinc surface condition.** More time/heavier chemistry → heavier, darker film. But there's a ceiling — **over-immersion** stops building protective film and starts powdering it. Bath control (pH and chemistry, by your lab) keeps color repeatable run-to-run.

THE ONE RULE OPERATORS BREAK MOST

THE DON'T-OVER-DRY RULE

DRYING, AGING, AND THE SOFT GEL FILM — WHY HEAT IS THE ENEMY HERE

This deserves its own page because it's the most common way good chromate parts get ruined. **The fresh chromate is a hydrated gel.** Two things must happen, gently:

1. **Drying** — remove surface water so parts can be handled/packed.
2. **Aging/curing** — over the next several hours to ~24 h, the gel slowly dehydrates and **hardens** into a coherent, durable film at room temperature.

Heat short-circuits this badly. Drying too hot (or rinsing too hot) **flash-dehydrates the gel**, causing it to **shrink, craze (microcrack), and lose its bound water and mobile Cr^{6+}** — destroying corrosion resistance and self-healing. The killer is that the damage is usually **invisible**: the part looks fine and fails salt spray.

DO

- Dry **below the TDS limit** (commonly ≤ ~60°C / 140°F).
- Use **warm air, spin-dry, or ambient.**
- Let parts **age/cure** ~24 h before rough handling.
- Trust **color + rub test + salt spray.**

DON'T

- **Bake** in a hot oven like a phosphate/paint part.
- Use a **hot final rinse** to "speed drying."
- Heavily **stack/abrade** fresh, soft parts.
- Assume "looks dry, looks fine" = good.

★ REMEMBER

Cool dip, cool rinse, gentle low-temp dry, then let it age. If you remember one *process* rule from this whole manual, make it this: **never over-dry a fresh chromate.** Heat is the enemy of a gel coating.

☀ FUN FACT

This is exactly backwards from the iron-phosphate manual, where a 250–400°F dry-off oven is mandatory and *good*. Same company, same conveyor maybe — opposite rule. It's a perfect example of why you run *each* chemistry by *its own* TDS and never carry a habit from one line to another.

WHY HEX CHROMATE EARNED ITS REPUTATION

THE SELF-HEALING MECHANISM

ACTIVE CORROSION PROTECTION — THE PROPERTY, AND THE PROBLEM, ARE THE SAME MOLECULE

The mechanism, step by step:

1. The cured film is a chromium/zinc oxide-hydroxide matrix that **holds a reservoir of soluble hexavalent chromium.**
2. When the film is **scratched** down to bare zinc and the spot gets **wet**, the trapped Cr^{6+} **dissolves into the moisture** at the damage.
3. That mobile Cr^{6+} **migrates to the exposed zinc** and **re-passivates it** — chemically forming fresh chromate over the bare metal.
4. The scratch is **re-protected**, without any operator action.

This is called **active** corrosion protection (the coating responds to damage), versus **passive** barrier protection (paint just sits there).

★ **REMEMBER**

Self-healing = mobile Cr^{6+} leaching to re-protect damage. It's why hex chromate gives such strong bare-metal and scratched-surface protection so cheaply — and why **trivalent passivates, which lack that mobile Cr^{6+}, generally need a sealer/topcoat to approach the same performance.**

⚙ **TECHNICAL STUFF**

The same mobility that heals scratches is the same mobility that makes the chromium **bioavailable and hazardous** — and the same reason **over-rinsing or over-drying leaches it out** and kills the property. Everything about hex chromate traces back to this one fact: *the healing agent is soluble Cr(VI).* That's the property, the hazard, and the regulatory problem, all in one.

⚠ **HEADS-UP**

Because self-healing depends on the Cr^{6+} staying in the film, the upstream lessons all connect: **don't over-rinse (Station 05), don't over-dry (Station 07).** Strip the reservoir and you keep the carcinogen risk while losing the benefit — the worst of both worlds.

THE DECISION DRIVING THE INDUSTRY

HEX VS. TRIVALENT

WHY THE WORLD MOVED TO CR(III) — AND WHERE CR(VI) STILL HANGS ON

FACTOR	HEXAVALENT (CR VI)	TRIVALENT (CR III)
Toxicity	**Carcinogen**	Far less hazardous (trace nutrient form)
Self-healing	**Strong** (mobile Cr^{6+})	Weak; **usually needs topcoat/seal**
Corrosion-per-cost	**Excellent**, cheap	Good; topcoats add cost/steps
Color range	Clear, yellow, olive, black (classic)	Clear/blue and black common; yellow/iridescent harder, often via topcoats
RoHS / ELV / REACH	**Restricted**	**Compliant** — modern standard
Where used	Phasing out; survives in **some military/defense and industrial** where RoHS/ELV don't apply	**Default** for new commercial/automotive/electronics

THE REGULATORY DECISION TREE (PLAIN VERSION)

- **Is the part for electronics (RoHS), automotive (ELV), or an EU/REACH-governed market? → Trivalent (or other Cr-free).** Hex is not acceptable.
- **Is it for an application that still permits hex (some defense/industrial)?** → Hex may be used, with full Cr(VI) controls — but confirm; exemptions narrow over time.
- **When in doubt?** → Treat hex as restricted and **confirm the customer spec and current regulations.**

★ **REMEMBER**

The honest summary: **hex chromate is cheaper and self-healing and gives the best classic corrosion-per-dollar — but it's a carcinogen and RoHS/ELV/REACH-restricted, so it's being phased out.** Trivalent is the RoHS-compliant modern standard that usually needs a seal/topcoat to match hex. **We build the trivalent manual next** — it shares this exact chromate-conversion foundation with a Cr(III) chemistry and a mandatory-seal mindset.

⚠ **HEADS-UP**

Don't let "hex is being phased out" make you complacent about the tank you're running *today.* If your shop runs hex (for a market that still allows it), it is still a carcinogen and still demands every Cr(VI) control in this manual.

KEEPING CR(VI) OUT OF THE ENVIRONMENT

CR(VI) WASTE TREATMENT

REDUCE, THEN PRECIPITATE — CHROMIUM NEVER GOES DOWN A DRAIN

Every drop of Cr(VI) — drag-out, rinse water, dumped baths, spills — must be treated before discharge. The classic two-step:

1. **Reduction ($Cr^{6+} \rightarrow Cr^{3+}$).** The wastewater is **acidified** (low pH) and a **reducing agent** (e.g., sodium metabisulfite/sulfite, or ferrous sulfate) converts the dangerous, soluble **hexavalent** chromium into far-less-hazardous **trivalent** chromium. You can often *see* it: the yellow Cr^{6+} shifts toward green/blue Cr^{3+}.
2. **Precipitation / removal (Cr^{3+} out of solution).** The pH is then **raised** (neutralized) so the trivalent chromium **precipitates as chromium hydroxide**, which is settled/filtered out as sludge for proper hazardous-waste disposal. The treated water is tested and discharged only within permit limits.

★ **REMEMBER**

Reduce then precipitate: $Cr^{6+} \rightarrow Cr^{3+} \rightarrow$ chromium-hydroxide sludge. The hazardous hexavalent form is destroyed first; only then is the chromium pulled out of the water. **Cr(VI) never, ever goes down a drain untreated** — that's both illegal and an exposure hazard.

⚙ **TECHNICAL STUFF**

The reduction step needs **low pH** to work efficiently; the precipitation step needs **near-neutral/slightly alkaline pH**. That's why a chrome waste-treatment system is essentially a controlled pH-and-redox process — exactly the same reduce-then-precipitate logic used to treat hard-chrome plating waste. The end sludge is a regulated hazardous waste with its own manifest/disposal chain.

⚠ **HEADS-UP — AIR SIDE TOO**

The tank exhaust carries Cr(VI) mist. A **scrubber** washes it out of the air before discharge, and the operation runs under **air permits.** If the scrubber or exhaust is down, the tank isn't compliant *or* safe — stop and report it.

☀ **FUN FACT**

The color change during reduction (yellow → green/blue) is the same chemistry, run in reverse, that colors your parts: in the tank, zinc reduces a little Cr^{6+} to Cr^{3+} to build the film; in waste treatment, a reducing agent reduces *all* of it to Cr^{3+} to destroy the hazard. Color is a clue in both places — but never a substitute for proper testing.

GETTING MORE FROM THE FILM

SEALS & TOPCOATS

OPTIONAL ON HEX, OFTEN MANDATORY ON TRIVALENT

A **seal** or **topcoat** is a thin layer over the rinsed chromate that closes pores and adds a barrier, corrosion life, color, or lubricity.

TYPE	ADDS	CR(VI)?	NOTES
Non-chrome organic/polymer topcoat	Corrosion life, color, **torque-tension control** on fasteners	No	Increasingly the default
Silicate / inorganic seal	Barrier, corrosion life	No	Common chrome-free seal
Chromic/chromate seal	Extra protection	**Yes**	Full Cr(VI) hazards/controls

• WHY FASTENERS CARE ABOUT TOPCOATS

A topcoat sets the **coefficient of friction**, which controls the **torque-tension** relationship when bolts are tightened. ASTM **F1941** jobs often specify this. A bare chromate can be too “grabby” or too slick; the topcoat dials it in.

★ **REMEMBER**

On **hex** chromate a seal is usually optional polish; on **trivalent** it’s often **required** to reach the spec. If you move to the trivalent line later, expect the seal/topcoat to become a non-negotiable station, not an add-on.

⚠ **HEADS-UP**

A chromate seal is **still Cr(VI).** Many shops choose non-chrome seals precisely to avoid a second carcinogen tank. Know which one your shop runs and apply the matching controls.

⚙ **TECHNICAL STUFF**

A seal/topcoat works by closing the slightly porous, chemically “open” chromate film and adding a final barrier layer. On hex chromate, the self-healing film already performs well, so a seal is a corrosion/lubricity bonus. On trivalent, the seal is doing real load-bearing work to make up for the missing mobile-Cr(VI) self-healing — which is why it’s usually mandatory there.

HOW WE PROVE IT WORKED

QUALITY CHECKS

A CHROMATE IS JUDGED BY CORROSION LIFE AND FILM INTEGRITY, NOT JUST BY LOOKING SHINY

CHECK	WHAT IT TELLS YOU	HOW
Color / Type	Right finish; rough protection level	Visual vs. standard (clear/yellow/olive/black)
Coverage / uniformity	No bare/thin spots	Visual, all faces
Rub / no-powder test	Film cured and adherent (not over-dipped/over-dried)	Light wipe — color must not rub off as powder
Salt spray (white rust)	Under-finish corrosion life	**ASTM B117** salt-fog; hours to white rust per spec
Adhesion (if painted)	Paint grips the film	**ASTM D3359** crosshatch/tape on painted samples
Coating presence/weight	Film actually formed	Lab methods; presence tests per spec

★

REMEMBER

The real verdict is **salt spray (ASTM B117)** — hours to **white rust** (zinc corrosion) and then to **red rust** (steel) per the customer spec. Color and the rub test are your fast **in-process predictors**; salt spray is the final judge. If salt spray fails but color looked fine, suspect **over-rinse or over-dry** stripping the film.

⚙

TECHNICAL STUFF — WHAT “SALT SPRAY” IS

Painted or bare samples sit in a sealed cabinet sprayed with salt fog (ASTM B117); you record the *hours* until white rust (zinc) or red rust (steel) appears. More hours = better protection. It’s how customers specify and verify corrosion requirements — and how you’d qualify a color/Type/seal combination.

💡

TIP

Keep retained samples and a **color standard** at inspection. Run a periodic salt-spray on production samples so you catch a drifting bath, a creeping rinse temperature, or a too-hot dryer *before* a customer does.

FIELD REFERENCE

TROUBLESHOOTING QUICK-INDEX

FIND YOUR SYMPTOM, READ THE LIKELY CAUSE, TRY THE FIX

SYMPTOM	LIKELY CAUSE(S)	FIRST THINGS TO CHECK
No / weak color	Dip too short/weak; bath cold or out of pH; **stale or poorly activated zinc**	Lengthen dip to target color; check pH/chemistry; re-activate; confirm zinc freshness
Powdery / chalky / rub-off film	**Over-immersion**; bath too aggressive; pH off; **over-dried**	Shorten dip; verify pH/temp per TDS; lower dryer temp
Poor salt-spray (looks fine, fails test)	**Over-rinse or over-dry stripped the film**; thin film; wrong Type for spec	Cool/shorten rinse; **lower dryer temp**; run heavier color/seal per spec
Dried-out / crazed / cracked film	**Hot rinse or hot dry** dehydrated the gel	**Drop dryer temp below TDS limit**; eliminate hot final rinse
Patchy / uneven color	Poor activation; nesting/overpacked barrels; uneven agitation	Re-activate; reload; check agitation
White rust appearing	Thin/under-protective chromate; damaged film; humidity in storage	Run correct Type/seal; protect cured parts; check drying
Faded yellow	Over-rinsed/over-leached (lost Cr^{6+})	Cool/shorten rinse; check bath
Color won't repeat run-to-run	Bath drifting (pH/chemistry/temp); zinc condition varies	Titrate/adjust per TDS (lab); standardize incoming zinc

💡 **TIP**

The fastest diagnostic instinct: **read the part backward through the line.** No color? Suspect activation and the chromate dip. Powdery? Suspect over-immersion or over-drying. Fails salt spray but looks good? Suspect a **hot rinse or hot dryer** that stripped the film. Crazed? Definitely heat. The defect usually names the stage that failed it.

⚠ **HEADS-UP**

Before you change bath chemistry, pH, temperature, or dryer settings, confirm with your lead and the supplier TDS. Bath adjustments and additions to a **Cr(VI)** tank are made only by authorized personnel, in full PPE, with ventilation and mist suppression running.

YOUR DECODER RING

GLOSSARY — EVERY TERM, PLAIN AND SIMPLE

ANYTIME A WORD IN THIS MANUAL MADE YOU PAUSE, FLIP BACK HERE

Activation / bright dip: A quick dilute-acid (often nitric) dip that strips oxide/smut off zinc and re-exposes fresh, reactive metal before chromating.
Activator / accelerator: Additives (sulfate, formate, fluoride, etc.) that control how the chromate attacks zinc and builds film. Per TDS.
Ambient: Room temperature; no heater.
ASTM B117: The salt-spray (salt-fog) corrosion test; results in *hours* to white/red rust.
ASTM B633: Spec for electrodeposited zinc on steel, including chromate **Types** and thickness **Classes**.
ASTM F1941: Spec for electrodeposited coatings on threaded fasteners.
Cadmium: A plated metal historically chromated like zinc; now restricted in many uses.
Chromate / passivate: The conversion coating formed on plated zinc by an acidic chromium solution; protects, self-heals, and colors.
Class (B633): The *zinc thickness* / service condition (separate from Type).
Conversion coating: A film formed by chemically reacting (converting) the metal's own surface — no electric current.
Cr(VI) / hexavalent chromium: Chromium in the +6 state; a confirmed human carcinogen; the active, self-healing ingredient of classic chromates. OSHA 29 CFR 1910.1026.
Cr(III) / trivalent chromium: Chromium in the +3 state; far less hazardous; basis of RoHS-compliant passivates (next manual).
Drag-out / drag-in: Solution clinging to a part leaving a tank (drag-out) and contaminating the next (drag-in) — on a Cr(VI) line, a carcinogen to contain.
ELV: End-of-Life Vehicles directive — restricts Cr(VI) in automotive parts.
Iridescence: The rainbow/blue-to-gold color of a thin film (interference); part of how a chromate's color reads.
Olive-drab: The heaviest classic chromate color; highest classic protection; military.
PEL / Action Level: OSHA Cr(VI) air limits — 5 μg/m³ (PEL) and 2.5 μg/m³ (action level), 8-hr averages.
pH: Acidity/alkalinity scale (0 acid – 14 base; 7 neutral). Chromate baths run ~1.5–2.5.
Reduce-then-precipitate: Cr(VI) waste treatment — reduce $Cr^{6+} \rightarrow Cr^{3+}$ (low pH), then precipitate Cr^{3+} as hydroxide (raise pH).
REACH: EU regulation placing hexavalent-chromium substances under authorization/restriction.
RoHS: Restriction of Hazardous Substances — restricts Cr(VI) in electrical/electronic equipment.
Seal / topcoat: A thin layer over the chromate for extra corrosion life, color, or lubricity; optional on hex, often required on trivalent.
Self-healing (active protection): The film's ability to re-protect a scratch as soluble Cr^{6+} leaches over the bare zinc. The signature of hex chromate.
Substrate: The base surface being coated — here, electroplated zinc / zinc alloy / cadmium.
TDS / SDS: Technical Data Sheet (how to run a product) / Safety Data Sheet (its hazards). Follow both.
Type (B633): The *supplementary finish* (chromate) on the zinc — Type II colored, Type III colorless; verify exact numbers against the current revision.
White rust: Powdery white zinc-corrosion product; the first failure a chromate is meant to delay.
Zinc / zinc-alloy plating: The electrodeposited metal layer the chromate converts (zinc, zinc-nickel, zinc-iron, zinc-cobalt).

TEMPLATE — PRINT ONE PER OPERATOR

OPERATOR TRAINING MATRIX

HOW A SUPERVISOR PROVES — ON PAPER — THAT EACH OPERATOR IS TRAINED AND SIGNED OFF BEFORE RUNNING SOLO

★ **REMEMBER**

An operator is “line-qualified” only when every station they run is signed. **Nobody runs the hexavalent chromate tank, a chromate seal, or handles Cr(VI) waste solo until the Cr(VI) program / PPE / SDS row is signed.** Safety qualification comes first, always.

Sign-off levels: O = Observed only · **A** = Assisted under supervision · **Q** = Qualified to run solo · **T** = Qualified to Train others.

#	STATION / SKILL	O/A/Q/T	TRAINER INIT & DATE	OPERATOR INIT & DATE	RE-CERT DUE
1	**Cr(VI) program / PPE & SDS (all stations)**	____	________	________	______
2	Station 01 — Receive & Inspect (freshness, substrate)	____	________	________	______
3	Station 02 — Nitric Activation / Bright Dip	____	________	________	______
4	Station 03 — Rinse (post-activation)	____	________	________	______
5	Station 04 — Hexavalent Chromate (color/Type, ventilation/mist)	____	________	________	______
6	Bath control — pH / chemistry checks	____	________	________	______
7	Station 05 — Rinse (cool/controlled, no leaching)	____	________	________	______
8	Station 06 — Seal/Topcoat (identify chrome vs. non-chrome)	____	________	________	______
9	Station 07 — **Dry-Off (low-temp, don’t over-dry)**	____	________	________	______
10	Station 08 — Inspect & Hand-Off (color, rub test)	____	________	________	______
11	Self-healing & color-vs-protection understanding	____	________	________	______
12	Cr(VI) waste treatment (reduce → precipitate)	____	________	________	______
13	Drag-out control / no Cr(VI) to drain	____	________	________	______
14	LOTO (heaters/pumps/hoists/dryer/ventilation)	____	________	________	______
15	Emergency response — eyewash, shower, spill	____	________	________	______
16	Quality — salt spray / color standard / troubleshooting	____	________	________	______

Operator name: ______________________________ **Hire/start date:** __________________

Supervisor name: ______________________________ **Review cycle:** ☐ Annual ☐ Other: ____________

⚠ **HEADS-UP**

Safety rows (1, 12, 13, 14, 15) are not optional and not “fill in later.” An operator may not work *any* station — and especially not the Cr(VI) tank, a chromate seal, or waste treatment — until the Cr(VI)/PPE/SDS, waste, drag-out, LOTO, and emergency rows are signed. Safety competency comes before production competency, always.

PART FOUR — BACK MATTER & CERTIFICATION

20 QUESTIONS · PASSING SCORE 80% (16/20)

HEXAVALENT CHROMATE COMPLETION TEST

COVERS EVERY STATION PLUS SAFETY — ANSWER IN YOUR OWN WORDS WHERE ASKED

★ **REMEMBER**

Passing score: 80% (16 of 20 correct). Any **safety** question answered wrong (Q5, Q6, Q14, Q15) is an automatic fail of the safety gate — re-teach and re-test before sign-off, regardless of total score. The Answer Key follows — no peeking until you're done!

Name: ______________________ **Date:** ____________ **Score:** ________ / 20 **Safety-gate Qs (all correct): 5, 6, 14, 15**

Q1. A chromate conversion coating forms by:
A. An electric current depositing chromium from a bath onto the part B. A chemical reaction between the freshly plated zinc surface and the process solution (no electric current) C. Spraying molten chromium onto the zinc D. Baking a chromate powder onto the part in a hot oven

Q2. (Short answer) Explain what "self-healing" means for a hex chromate, and which **chromium species** provides it.

Q3. A hexavalent chromate bath typically runs at what **pH**?
A. ~1.5–2.5 (strongly acidic) B. ~4.5–5.5 (mildly acidic) C. 7.0 (neutral) D. 11–13 (alkaline)

Q4. (Short answer) Where does the chromate step fit in the overall process, and on which **substrates** is it used?

Q5. **[SAFETY GATE]** The classic chromate bath is best described as:
A. A confirmed human carcinogen (hexavalent chromium) regulated under OSHA 29 CFR 1910.1026 B. A harmless, food-safe rinse C. A non-hazardous trivalent solution D. An alkaline degreaser

Q6. **[SAFETY GATE]** OSHA's permissible exposure limit (PEL) for hexavalent chromium in air (8-hr average) is:
A. 500 µg/m³ B. 50 µg/m³ C. 5 µg/m³ D. 5 g/m³

Q7. (Short answer) Why must you **NOT dry a fresh chromate film at high temperature**? What does over-drying do to the film?

Q8. Which classic chromate color provides the **least** bare-zinc corrosion protection?
A. Olive-drab B. Clear / blue C. Yellow / gold iridescent D. Black

Q9. (Short answer) Where does a chromate's **color** come from, and what does going clear → yellow → olive-drab tell you about protection?

Q10. Over-drying or over-baking a fresh chromate typically:
A. Makes it shinier and tougher B. Has no effect C. Cracks/dehydrates the soft gel film and destroys self-healing and corrosion resistance D. Turns it into trivalent chromium

Q11. (Short answer) Why must the **post-chromate rinse** be *cool and brief*? What happens if it's hot or prolonged?

Q12. A hex chromate's self-healing comes from:
A. Soluble hexavalent chromium (Cr^{6+}) in the film leaching out to re-passivate a scratch   B. A nickel underlayer   C. Baked-on epoxy paint   D. The steel substrate rusting protectively

Q13. Which regulations are the main reason industry is phasing out hexavalent chromate in favor of trivalent passivates?
A. FDA food-contact rules   B. FAA flight rules   C. Building fire codes   D. RoHS, ELV, and REACH (they restrict hexavalent chromium)

Q14. (Short answer) **[SAFETY GATE]** List **four** pieces of PPE you must wear at the hexavalent chromate station, and name **one engineering control** that is your *primary* protection against the mist.

Q15. **[SAFETY GATE]** The correct waste treatment for hexavalent-chromium rinse water/spills is to:
A. Pour it straight down the drain — it's dilute   B. Reduce Cr^{6+} to Cr^{3+} (at low pH), then raise pH to precipitate the chromium as hydroxide for disposal   C. Mix it with the alkaline cleaner and discharge   D. Let it evaporate in an open tank

Q16. The modern, RoHS-compliant replacement for hexavalent chromate is:
A. Hot-dip galvanizing   B. Iron phosphate   C. A trivalent (Cr III) passivate (usually with a seal/topcoat)   D. Bare bright zinc, unpassivated

Q17. (Short answer) Give **two** honest tradeoffs between hexavalent and trivalent chromate (one advantage of each is fine).

Q18. The everyday spec covering electrodeposited **zinc on steel plus chromate Types/Classes** is:
A. ASTM B117   B. MIL-DTL-5541   C. ASTM D3359   D. ASTM B633

Q19. A chromate conversion coating is applied to:
A. Bare carbon steel directly   B. Anodized aluminum   C. Cured powder coat   D. Freshly electroplated **zinc / zinc-alloy (or cadmium)** surfaces

⚠ **SAFETY SECTION — MUST BE CORRECT TO CERTIFY**
Questions 5, 6, 14, and 15 cover hazards that can permanently harm you or others. A miss on any safety item means re-teach and re-test before sign-off.

Q20. (Short answer) Name **two** quality checks used to verify a chromate finish and, in a few words, what each tells you.

End of test. Hand to your trainer for grading.

FOR TRAINERS & SELF-CHECKERS

ANSWER KEY

SHORT-ANSWER RESPONSES DON'T NEED TO MATCH WORD-FOR-WORD — LOOK FOR THE KEY CONCEPTS IN BOLD

> **HEADS-UP**
> Keep this page out of the trainee's copy. Questions **5, 6, 14, and 15** are safety-critical — any wrong answer there requires re-training on that topic before sign-off, regardless of total score.

Self-check answer distribution (the 12 multiple-choice items): A = 3, B = 3, C = 3, D = 3. If your key clusters on one letter, you've mis-keyed.

Q1. B — A chemical reaction between the zinc surface and the solution; **no electric current**.

Q2. When the film is **scratched and gets wet**, **soluble hexavalent chromium (Cr^{6+})** trapped in the film **leaches out and re-passivates the exposed zinc** — the coating patches its own scratches ("active" protection). Credit for "trapped Cr^{6+} leaches to heal damage."

Q3. A — ~1.5–2.5 (strongly acidic).

Q4. It's the **finishing/passivating step after zinc plating** (the "passivate" the zinc line hands off to). Used on **electroplated zinc, zinc alloys (zinc-nickel/iron/cobalt), and cadmium** — *not* bare steel or aluminum.

Q5. A — A confirmed human carcinogen (hexavalent chromium); OSHA 29 CFR 1910.1026.

Q6. C — 5 µg/m³ (action level 2.5 µg/m³).

Q7. Because the fresh film is a **soft hydrated gel**; high heat **flash-dehydrates and cracks/crazes it, drives off the soluble Cr^{6+}, and destroys corrosion resistance and self-healing** — often invisibly (looks fine, fails salt spray). Keep dry-off below the TDS limit (commonly ≤ ~60°C/140°F).

Q8. B — Clear / blue (lightest film, least protection). Olive-drab is the most protective of the classics.

Q9. Color comes from **film thickness + trapped Cr(VI) + additives/dyes.** Clear → yellow → olive-drab means **heavier film, more chromium, and more corrosion protection** (clear = least, olive = most).

Q10. C — Cracks/dehydrates the gel and destroys self-healing/corrosion resistance.

Q11. Because the **fresh film is a soluble gel** holding the self-healing Cr^{6+} reservoir. A **hot or prolonged rinse leaches the chromium back out**, fading color and weakening protection. Rinse cool and brief — enough to remove drag-out, not strip the film.

Q12. A — Soluble hexavalent chromium leaching out to re-passivate a scratch.

Q13. D — RoHS, ELV, and REACH restrict hexavalent chromium.

Q14. PPE (any four): sealed chemical splash goggles, face shield, chromic-acid-rated chemical gloves, chemical apron/suit, chemical-resistant/non-slip boots, respirator per the Cr(VI) program. **Primary engineering control (any one):** local exhaust ventilation, fume/mist suppressant on the bath surface (and a scrubber). (Engineering controls are primary; PPE/respirator are last.)

Q15. B — Reduce $Cr^{6+} \rightarrow Cr^{3+}$ at low pH, then raise pH to precipitate chromium hydroxide for disposal. Cr(VI) never goes down a drain.

Q16. C — A trivalent (Cr III) passivate, usually with a seal/topcoat.

Q17. Any two, e.g.: **Hex advantages** — cheaper, strong self-healing, best classic corrosion-per-cost. **Hex disadvantages** — carcinogen, RoHS/ELV/REACH-restricted. **Trivalent advantages** — RoHS-compliant, far safer. **Trivalent disadvantages** — weaker self-healing, usually needs a seal/topcoat, added cost/steps.

Q18. D — ASTM B633 (MIL-DTL-5541 is the *aluminum* chem-film spec; B117 is salt spray; D3359 is paint adhesion).

Q19. D — Freshly electroplated zinc/zinc-alloy (or cadmium) surfaces.

Q20. Any two of: **color/Type vs. standard** (right finish & rough protection); **rub/no-powder test** (cured, adherent film, not over-dipped/over-dried); **salt spray ASTM B117** (hours to white/red rust = corrosion life); **coverage/uniformity** (no bare spots); **paint adhesion ASTM D3359** if painted.

> **REMEMBER**
> 16/20 passes — but every safety question (5, 6, 14, 15) must be correct to certify. Miss a safety item → re-teach and re-test before sign-off. We don't gamble with people — especially on a Cr(VI) line.

PRINT, FILL IN, SIGN

PLATING POSTERS

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CERTIFICATE OF COMPLETION

CHROMATE CONVERSION COATING (HEXAVALENT) TRAINING PROGRAM

This certifies that

OPERATOR NAME

has successfully completed the **Hexavalent Chromate Conversion Coating** training course — covering all eight process stations (Receive & Inspect, Nitric Activation/Bright Dip, Rinse, Hexavalent Chromate Passivate, Controlled Rinse, Seal/Topcoat, Low-Temperature Dry-Off, and Inspect/Hand-Off), the fundamentals of chemical conversion coating without electric current, the self-healing mechanism of hexavalent chromate, the clear/yellow/olive-drab/black color palette and how color signals protection, the don't-over-dry rule, hex-vs-trivalent positioning and the RoHS/ELV/REACH regulatory driver, Cr(VI) waste treatment (reduce-then-precipitate), troubleshooting, and the safety practices for a **hexavalent-chromium (Cr VI) carcinogen** line — including ventilation, mist suppression, respiratory protection, hygiene, drag-out control, and emergency response — and has passed the Completion Test with a score of ________ / 20, including all required safety competencies.

This operator is hereby cleared for **independent work** at the certified stations listed on the Operator Training Matrix.

Date of completion: ____________________ **Re-certification due:** ____________________

TRAINEE SIGNATURE

CERTIFYING TRAINER

SUPERVISOR

Training reference only. Verify all parameters against your chemistry supplier's TDS, customer specifications, and applicable regulatory requirements before production use. Hexavalent chromium is a confirmed human carcinogen and is restricted under RoHS, ELV, and REACH — always consult current SDS and comply with OSHA 29 CFR 1910.1026, EPA, and local regulations.